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Context-aware AR adaptive information push for product assembly: Aligning information load with human cognitive abilities

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ABSTRACT

Augmented Reality (AR) innovates product assembly by overlaying intuitive visual instructions onto the physical workspace, improving operational efficiency. However, existing AR-guided assembly methods has the following drawbacks. Most studies lack consideration of the dynamic adaptation of operators' cognitive abilities. Excessive or insufficient information push may cause cognitive overload or information underutilization, which has instead a negative impact on assembly work. In addition, due to lack of automatic perceiving assembly state, frequent human-interface interactions interferences the workflow continuity, which is particularly disadvantageous for manual assembly work. To address the problems, we propose an intelligent AR-guided adaptive information push (AR-AIP) method based on context-awareness, which personalizes information push according to individual cognitive abilities. Firstly, the method comprehensively assesses human cognitive abilities by separately considering factors such as fatigue state and skill level. Secondly, it quantifies the information load associated with various visual representations of AR content, aiming to identify the most suitable combination that aligns with each user's cognitive abilities. Thirdly, the perception of the assembly process and its state is fundamental for achieving adaptive content push. This study, therefore, employs context awareness by integrating the recognition of assembly actions and the detection of parts, facilitating the automatic inference of the assembly process and its state. The AR-AIP aims at three "rights", i.e., push the right information in the right AR representation forms to the operator at the right time, proactively and automatically, with minimal manual interactions. This approach is designed to alleviate the cognitive burden on operators, thereby enhancing assembly quality and efficiency. A comparative study on electronic equipment assembly shows that compared to traditional AR methods, AR-AIP significantly improves task completion time, reduces errors, lowers cognitive load, enhances user experience while minimizing skill and fatigue variations. The research findings offer insights for designing AR visual content to assist assembly tasks, providing a new approach for proactive perception and understanding of real-time assembly states.

1. Introduction

For knowledge-intensive assembly tasks, relying on traditional manuals imposes significant cognitive burdens on operators [1,2]. The advent of augmented reality (AR) technology has innovated assembly by enabling real-time interaction between physical objects, virtual data, and human, making it easier to integrate physical and digital elements [3]. AR overlays visual aids such as text, holograms, and markers onto the real-world setting, facilitating a more intuitive understanding of complex assembly processes [4,5].

Assembly process information is crucial for on-site operators, and thus the effectiveness and user-friendliness of its presentation significantly influence assembly operation efficiency and quality [6]. Despite AR's various information representations, their efficacy can vary among users, mainly due to insufficient attention on the interaction between information load and operators' cognitive abilities [7,8]. Cognitive ability refers to the capacity of on-site assembly operators to correctly understand assembly process information in AR-based visualizations. Information load refers to the cognitive effort required to reduce the uncertainty in assembly process information, which is quantified using

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Abbreviations and Nomenclature

<i>AI</i>	Artificial Intelligence
<i>ANN</i>	Artificial Neural Network
<i>ANOVA</i>	Analysis of Variance
<i>AP</i>	Average Precision
<i>ARDT</i>	Action Recognition Dataset
<i>AR</i>	Augmented Reality
<i>AR-AIP</i>	AR Adaptive Information Push method
<i>AR-Base</i>	Traditional AR-guided assembly method
<i>AUI</i>	Adaptive User Interfaces
<i>CAPP</i>	Computer-Aided Process Planning
<i>CLT</i>	Cognitive Load Theory
<i>CWT</i>	Continuous Wavelet Transform
<i>GPU</i>	Graphics Processing Unit
<i>HCI</i>	Human–computer Interaction
<i>IoU</i>	Intersection over Union
<i>LR</i>	Logistic Regression
<i>LRM</i>	Long Reach Multimode
<i>mAP</i>	mean Average Precision
<i>MFI-20</i>	Multidimensional Fatigue Inventory
<i>mRMR</i>	Maximum Relevance Minimum Redundancy
<i>NASA-TLX</i>	NASA Task Load Index
<i>NoE</i>	Number of errors
<i>PDDT</i>	Part Detection Dataset
<i>RF</i>	Random Forest
<i>SMA</i>	SubMiniature version A
<i>SOTA</i>	State-of-the-art
<i>SUS</i>	System Usability Scale
<i>SVM</i>	Support Vector Machine
<i>TCT</i>	Task Completion Time
<i>TSN</i>	Temporal Segment Network
<i>YOLO</i>	You Only Look Once, here refer to the deep learning model

information entropy [9–11]. The uncertainty in process information arises when AR presents it, as operators engage in perception, understanding, and linking it to the real world. Misalignment between information demand and cognitive ability can lead to overloading or underutilization, impacting guidance effectiveness [12,13]. Current AR-guided assembly systems often fail to adapt dynamically and proactively to the cognitive ability of operators due to inadequate real-time monitoring and responsive feedback. This deficiency hinders optimal performance, especially in balancing cognitive demands across various assembly tasks. The above disadvantages impede the widely acceptance of AR-guided assembly by industry.

AR has the potential to enhance operators' cognitive abilities by providing just-in-time information that aligns with their current cognitive state. However, this requires a nuanced understanding of how different AR presentation methods interact with various cognitive factors, such as skill level, and fatigue. For instance, operators with lower skill levels might benefit more from abundant visual guidance, while experienced operators may prefer condensed information that supports quick decision-making. Hence, establishing an effective information push method to ensure that the assembly process information matches the operators' cognitive abilities is of critical importance.

In addition, traditional AR-guided assembly methods (denoted as AR-Base) typically present assembly process information, such as text, images, videos, and 3D models, in a step-by-step visual format based on the assembly sequence. Operators are guided through the assembly process by interacting with buttons on the interface. AR-Base relies heavily on active interaction from the operator, such as clicking buttons or using voice commands. This explicit interaction can frequently disrupt the continuity of the operator's workflow [14,15]. This disruption can reduce the efficiency and effectiveness of AR technology in production

environments [16,17]. By automatically perceiving the current assembly state, AR systems can proactively interact with operators, pushing timely information without manual prompts [18,19]. This proactive information push not only ensures timely guidance but also maximizes operator productivity and minimizes disruptions [16,20].

Therefore, an AR-guided adaptive information push method (AR-AIP) is proposed from the perspectives of cognition and interaction proactivity. First, we assess cognitive abilities based on the two factors of skill and fatigue. Next, information load of the AR assembly process content is quantified using information entropy. To push information to the operator at the appropriate time, the assembly state needs to be acquired. In this paper, we achieve this by combining action recognition and part detection for inference. Finally, matching rules between information load and cognitive ability are established to ensure that the AR content presented to the operator is both reasonable and appropriate.

Compared to the AR-Base, AR-AIP focuses on how AR systems can perceive automatically the assembly state, on which basis, proactively and implicitly push the different guidance information for operators with different cognitive abilities. The specific differences are shown in the Table 1. AR-AIP offers (1) cognitive ability assessment considering operator fatigue and skill, (2) quantification of AR process information load, and (3) real-time assembly state inference based on context awareness, (4) alignment of information load to cognitive ability. To demonstrate the benefits of the proposed AR-AIP method, we conducted comparative evaluations against a traditional AR-guided assembly method (denoted as AR-Base) as a baseline. This involved gathering user data during the experiments for comprehensive analysis. To demonstrate the advantages of the proposed AR-AIP method, AR-Base is used as the baseline, and a comparative evaluation is conducted

Table 1
Differences in cognition and interaction between AR-AIP and AR-Base.

Dimensions	AR-Base (Traditional method)	AR-AIP (Proposed method in this paper)
Cognition	Diversified assembly process information will have a positive assistive effect on workers, but overlooks the consideration of workers' cognitive abilities and the negative impact of information load on operators.	It takes into account the cognitive load from diversified assembly process information and the adaptability of operators' cognitive abilities to the assembly information. It can quantify the information load of AR presentations and adaptively present assembly information based on operators' cognitive abilities (including skill and fatigue factors).
Interaction	Human active interaction: Human actively interact with the AR system through button clicks, voice commands, etc., to obtain assembly guidance information, disrupting the continuity of manual tasks.	AR system active interaction: The system perceives the assembly context (including part information and assembly actions) to infer the assembly state, proactively pushing relevant assembly guidance information to the operator, freeing their hands, and enabling immersive operation.

focusing on electronic equipment assembly tasks. The experimental process is designed to include the recruitment of qualified participants with varying skill levels and fatigue states, ensuring comprehensive and reliable results. During the experiment, detailed user data is collected and analyzed from several key perspectives, including task completion time, number of errors, cognitive load, system usability, and user experience. This approach allows for a more accurate comparison of the performance differences between the AR-AIP method and AR-Base.

The rest of the paper is structured as follows. Section 2 summarizes knowledge gaps in the literature. Then the AR-AIP framework, including the compositions and the working mechanism of AR-AIP is outlined in Section 3. The four main components of AR-AIP are introduced in detail, including cognitive ability assessment of operators in Section 4.1, assembly process information load evaluation in Section 4.2, context awareness-based assembly state inference in Section 4.3, and the aligning of information load to cognitive ability in Section 4.4. Section 5 verifies the effectiveness of our method via experiments and user study. Finally, Section 6 summarizes the study and proposes suggestions for future work.

2. Related works

2.1. AR's adaptability on operator cognitive ability

During the assembly process, operators rely on multiple senses, including vision, touch, and hearing, to gather information, perceive characteristics such as the shape, size, and texture of components, understand the assembly task's requirements and steps, and remember relevant operational protocols. An operator's cognitive state directly impacts work efficiency and quality. If tasks are overly complex or information is overwhelming, it can lead to cognitive fatigue and errors. Research has shown that AR technology can enhance operators' cognitive abilities by providing and visualizing critical information during the assembly process [21,22].

Cognitive abilities vary among individuals due to differences in prior knowledge, experience, and fatigue levels [23]. An individual's prior knowledge and experience reflect differences in skill levels [24]. Nishida et al. [25] demonstrated that AR can support the transfer of operational skills, helping individuals with lower skill levels to rapidly acquire new skills. Yan et al. [26] found that highly skilled operators encounter fewer difficulties in perceiving information, demonstrating stronger cognitive abilities. Their user study on AR-assisted assembly for operators with varying skill levels showed that AR can significantly enhance the cognitive abilities of novices.

On the other hand, fatigue can diminish an operator's cognitive abilities, reducing work efficiency and decision-making quality [27]. With advancements in brain-computer interface (BCI) technology, it is becoming increasingly feasible to monitor operators' fatigue levels in real-time [28–30]. Most studies monitor fatigue by collecting brainwave data during the task. For example, Liu et al. [31] used BCI monitoring to assess operator' fatigue state, dynamically adjusting task allocation in multi-user collaborative assembly. Cai et al. [32] incorporated fatigue monitoring as a key metric in human–robot collaborative task allocation research.

In summary, while existing studies have demonstrated the significant impact of AR on operators' cognitive abilities and proposed methods for monitoring fatigue using BCI, there remains a gap in research on how to comprehensively account for both skill levels and fatigue states, and how to adjust AR systems to accommodate cognitive changes in individuals. This paper will examine how different AR presentation methods can be tailored to address the dynamic shifts in cognitive abilities that occur due to the information load inherent in the assembly process.

2.2. Visualization of assembly process information

Information transfer during product assembly is central to the assembly activity, and AR technology provides a dynamic visualization method that surpasses traditional paper-based or electronic process documents [33]. AR enables the sequential display of discrete process information synchronized with assembly progress, task distribution, and material readiness, projecting content directly into the workspace. Li et al. [34,35] analyzed a vast number of assembly process documents and established a basic assembly process information model, achieving the visualization of basic assembly process information through AR. While Liu et al. [31] extended this to large-scale deployable mechanisms, integrating user, equipment, and part data into a cohesive AR-guided assembly system using HoloLens.

To design AR applications that better align with user cognitive habits, some studies have introduced cognitive rules into the visualization of AR assets and interfaces. Wang et al. [36] conducted research on visualizing AR assembly instruction information tailored to user cognition. They developed a semantic definition of assembly intent that aligns with user comprehension habits, allowing the transformation of advanced user experience into visual features and interaction methods suitable for novice training. Additionally, they addressed semantic expression issues of instructions at the comprehension level in AR information by proposing design rules and developing corresponding visual interaction interfaces [37].

However, despite advances in interactive interfaces and visual assets, challenges persist. Excessive information can lead to cognitive overload, and the reliance on human-initiated interactions often requires users to adapt to AR systems [38]. Additionally, frequent explicit interactions during assembly (such as button clicks and gesture operations) may cause operator fatigue, affecting task continuity and efficiency [38,39]. Most current research remains at the user experimentation stage, lacking AR visualization designs that consider the information load inherent in the assembly process. Therefore, quantifying and evaluating the specific load values generated by the assembly process and establishing a correlation with AR presentation forms are crucial for effectively presenting assembly process information.

2.3. Context awareness in assembly

Real-time monitoring of assembly progress is key to effectively coordinating production tasks and is also a prerequisite for proactive AR-guided assembly. This process is known as “context awareness”. In practical assembly operations, context awareness can be achieved

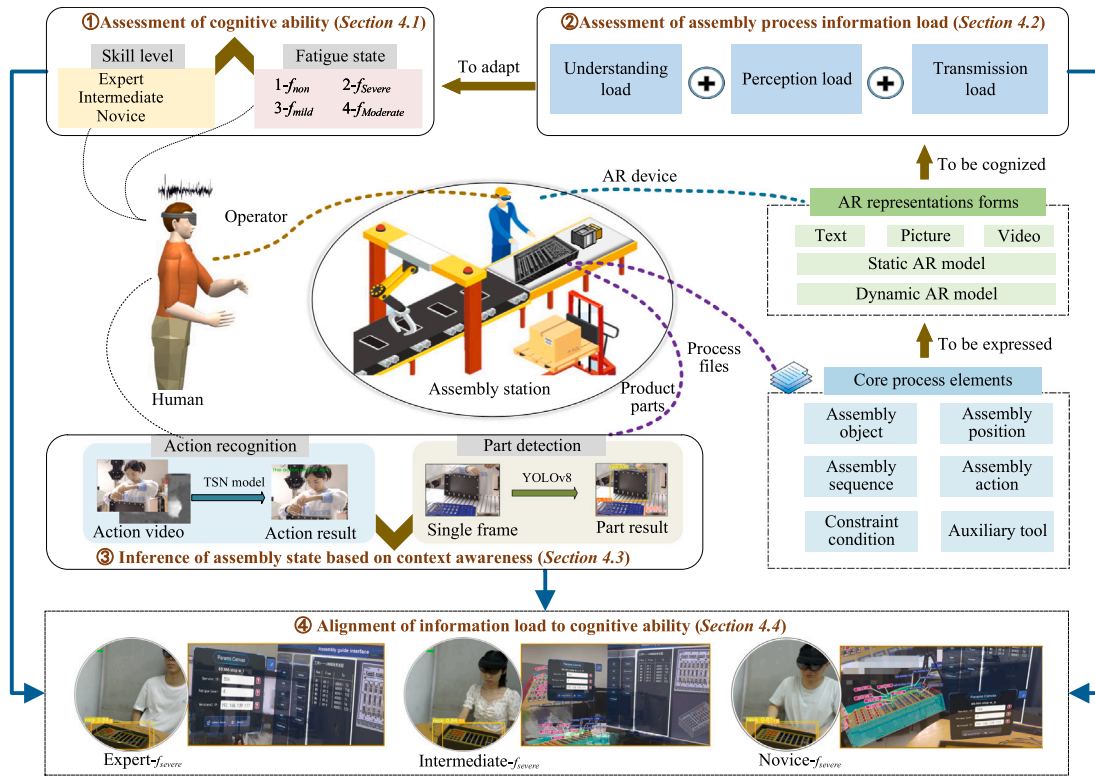


Fig. 1. The framework of AR-AIP.

by accurately detecting parts, assessing their positions and shapes, and inferring assembly sequences. Zheng et al. [40] showcased real-time monitoring of airplane cable assembly through deep learning and AR, offering operators clear visual guidance. Li et al. [41] identified assembly issues in aviation connectors, marking mismatches and triggering alerts. Fu et al. [42] developed an AR system that automatically initiates intuitive instructions based on assembly context. Fang et al. [43] developed a multimodal context-aware system that fuses point cloud recognition with image object detection to guide AR assembly. Similarly, Lv et al. [44] used a deep learning-based instance segmentation method to achieve pose estimation and state awareness of assembly objects, which can trigger AR instructions in real time during the assembly process.

Additionally, many studies have integrated human gesture and actions in assembly into context awareness. Tracking the operator's posture dynamics allows the system to deduce action intent and supply customized AR support for manual assembly. Dong et al. [45] focused on developing a dynamic gesture recognition system optimized for AR-assisted assembly, achieving action segmentation and standardizing assembly gestures. Yin et al. [46] emphasized implicit interaction AR systems that identify assembly stages by monitoring operator movements. Zhang et al. [47] established correlations between specific actions and target part detection, ensuring the system's deep understanding of the assembly context.

Through the discussion of the above work, it is clear that assembly context awareness plays a crucial role in enabling active interaction between AR systems and users [48]. This study is dedicated to deeply integrating this contextual information with AR-guided assembly. Specifically, it aims to recognize parts and actions during the assembly process, integrating the context-aware results into AR system, thus enabling intelligent inference of the assembly state. Moreover, by real-time sensing the operator's current fatigue state and skill level during AR-guided assembly, and intelligently adjusting the presentation of information based on calculated information load to match their cognitive abilities, it will help reduce cognitive burden and improve assembly quality and efficiency [20].

3. Framework of the proposed method

Fig. 1 outlines the AR-AIP framework. The AR-AIP process begins with assessing the operators' cognitive abilities, followed by quantifying the information load generated by AR in expressing the assembly process. It also involves monitoring the assembly procedures and its states, and finally establishing adaptive matching rules between cognitive ability and information load. The framework is designed to optimize the AR-guided assembly process by proactively pushing relevant process information to operators at the right time. It ensures that the pushing information is personalized aligning with the operator's cognitive ability. Therefore, the AR-AIP framework comprises four main components:

(1) Assessment of operators' cognitive ability (C_A). This involves gauging the operator's ability to comprehend assembly instructions in AR visualizations, taking into account their skill level and fatigue. Professional certifications determine skill level, whereas fatigue is assessed in real-time using machine learning algorithms through analyzing the brainwave signals. Details are provided in Section 4.1.

(2) Assessment of assembly process information load (H_{AR}). In the context of expressing the assembly process using AR representation, the information load is calculated across three stages: perception, understanding, and transmission. Each category of information load is quantified via the computation of information entropy. The detailed process is described in Section 4.2.

(3) Inference of assembly state based on context awareness. Context awareness in this study mainly refers to recognizing operator actions and detecting part types during the assembly process to infer the assembly procedure (M_k) and its state. Details are provided in Section 4.3.

(4) Alignment of information load (H_{AR}) to cognitive ability (C_A). After determining the current assembly state, the rules of aligning between the operator's cognitive ability and information load are established. Through the development of an adaptive information push mechanism, information will be pushed for the AR-guided assembly of the current procedure. Details are provided in Section 4.4.

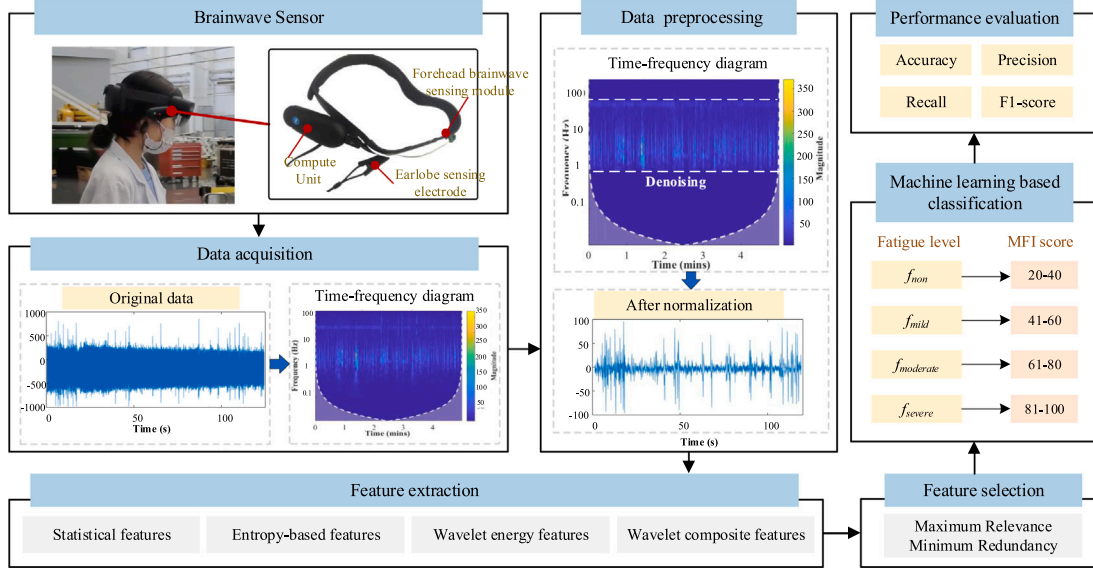


Fig. 2. The process of fatigue classification using brainwave signals.

Table 2
Classification of fatigue levels.

Category	Subjective scores	Fatigue levels
Non-fatigue (f_{non})	20–40	1
Mild-fatigue (f_{mild})	41–60	2
Moderate-fatigue ($f_{moderate}$)	61–80	3
Severe-fatigue (f_{severe})	81–100	4

4. Constituent components of AR-AIP

4.1. Assessment of operators' cognitive ability

Two factors, skill level and fatigue level, characterize the operator's cognitive ability (C_A):

$$C_A = \{skill, fatigue\} \quad (1)$$

Skill level determination typically requires rigorous assessment and considerable professional experience to obtain the corresponding certification, which remains unchanged for a considerable period. We categorize operator skill into three levels: expert, intermediate, and novice.

To determining fatigue levels, we followed a general evaluation method, that is, a combination of subjective assessment (questionnaire) and physiological signal detection [49]. Initially, the Multidimensional Fatigue Inventory (MFI) questionnaire survey [50] is utilized to establish fatigue level benchmarks for assembly operators. Based on the questionnaire scores, fatigue state is categorized into four levels, as shown in Table 2.

Brainwave signals, which are acquired using the Sichiray Brainwave Sensor, are used to assess the brain fatigue level. As shown in Fig. 2, the brainwave Sensor includes a sensing module and two earlobe sensing electrodes to capture brainwave signals from the forehead. The data can be transmitted to AR device via Bluetooth. The entire process is implemented in MATLAB.

Influenced by device limitations, noise, eye-blink artifacts, and transmission errors, raw brainwave signals need to be preprocessed, including steps of removing artifacts, removing outliers, and signal normalization. Brainwave rhythms are generally categorized into five

bands: δ wave (1–4 Hz), θ wave (4–8 Hz), α wave (8–12 Hz), β wave (12–30 Hz), and γ wave (30–100 Hz). Given that lower-frequency waves have a more substantial impact on individual fatigue, while high-frequency gamma waves are not a primary focus in fatigue studies, brainwave frequencies within the 1–70 Hz range are retained, with frequencies beyond this range being treated as noise [51,52]. Continuous Wavelet Transform (CWT) generates time–frequency diagram. Noise is eliminated by nullifying wavelet coefficients at designated frequencies. Subsequently, outliers are removed and the signals are normalized.

Feature extraction is then performed. Four types of features are selected, categorized into statistical, entropy-based, and wavelet energy features [53,54]. As shown in Table 3, some typical features are listed as candidates for fatigue classification. Statistical features are used to capture the variation in signal amplitude over time, while entropy features characterize the non-linear and non-stationary nature of brainwave signals. In the frequency domain, wavelet decomposition is used to obtain the five bands of brainwave rhythms. Subsequently, the wavelet energy and correlations across these different frequency bands are calculated.

The variance analysis and the Maximum Relevance Minimum Redundancy (mRMR) methods are used to select the most appropriate features [55]. First, the Pearson correlation coefficient is used to calculate the relevance $I(X_i; Y)$ between the feature X_i and the label Y . Then, the redundancy $I(X_i; X_j)$ between features is calculated to rank the features, and the top-ranked features are selected. The objective of mRMR is to select the optimal feature set, which involves balancing the relevance between features and the target variable (maximization), and minimizing the redundancy between features. It can be expressed as follows:

$$mRMR = \max_S \left[\frac{1}{|S|} \sum_{X_i \in S} I(X_i; Y) - \frac{1}{|S|^2} \sum_{X_i \in S} \sum_{X_j \in S} I(X_i; X_j) \right] \quad (2)$$

where, X_i and X_j are different features, Y is the label, and $|S|$ is the number of selected features. The relevance $I(X_i; Y)$ represents the correlation between the feature and the label. The redundancy $I(X_i; X_j)$ represents the redundancy between features, as given in Eq. (3), where $p(x_i, x_j)$ represents the joint probability distribution, and $p(x_i)$

Table 3
Brainwave signal features.

Statistical features	Entropy-based features	Wavelet energy features	Wavelet composite features
Mean value	Approximate entropy	Absolute energy of δ wave	$\delta/(\delta + \theta + \alpha + \beta + \gamma)$
Variance	Sample entropy	Absolute energy of θ wave	$\theta/(\delta + \theta + \alpha + \beta + \gamma)$
Standard deviation	Fuzzy entropy	Absolute energy of α wave	$\alpha/(\delta + \theta + \alpha + \beta + \gamma)$
Minimum value	Permutation entropy	Absolute energy of β wave	$\beta/(\delta + \theta + \alpha + \beta + \gamma)$
Maximum value	Entropy of mixing	Absolute energy of γ wave	Relative energy of $(\theta + \alpha)/\beta$

and $p(x_j)$ represents the edge probability density.

$$I(X_i; X_j) = \sum_{x_i \in X_i} \sum_{x_j \in X_j} p(x_i, x_j) \log_2 \left(\frac{p(x_i, x_j)}{p(x_i)p(x_j)} \right) \quad (3)$$

The brainwave features serve as the dataset. Given the limited data volume, to mitigate overfitting, we tend to choose structurally simplistic machine learning models rather than deep learning models. The four classification models – Support Vector Machine (SVM), Random Forest (RF), Logistic Regression (LR), and Artificial Neural Network (ANN) – were used to perform fatigue classification [56]. Finally, one of them will be selected after evaluating their performance.

4.2. Assessment of assembly process information load

Assembly involves operators constructing products from parts following a defined process, which includes various procedures. The process planner's knowledge is conveyed to the on-site operators through assembly process information. Then operators, based on their own understanding, conduct assembly operations using auxiliary tools such as fixtures, instruments, and equipment. On the basis of extensive research into assembly process documents for mechanical products, six types of core process elements (hereinafter referred to as core elements) are summarized, i.e., assembly object, assembly path, assembly action, assembly position, constraint condition, and auxiliary tool, which collectively express process information comprehensively. For a procedure M_k , its process information typically encompasses two or more of these six types of elements. The set of core elements can be represented as Eq. (4), where $I = 6$.

$$C = \{c_1, c_2, \dots, c_i \dots, c_I\} \quad (4)$$

Traditionally, assembly process information is recorded in assembly process documentation generated by Computer-Aided Process Planning (CAPP) systems. This documentation, which uses textual descriptions supported by illustrative drawings, is the most authoritative and accurate source of assembly process information. Taking the wire harnessing assembly in an electronic equipment as an example, the text description of the wiring harness connector assembly process is shown in Fig. 3.

Although the traditional way of presenting the assembly process in text has shortcoming such as abstract, lack of intuitiveness, difficult to comprehend, etc., their authority and detail are indispensable for operators. Therefore, when authoring processes content using AR technology, the original process text must be fully incorporated into the AR processes to ensure accuracy and professionalism. We define that AR representation forms include five categories: text (e_1), image (e_2), video (e_3), static AR model (e_4), and dynamic AR model (e_5). The set of AR representation forms can be represented as Eq. (5), where $J = 5$.

$$E = \{e_1, e_2, \dots, e_j \dots, e_J\} \quad (5)$$

The static AR model refers to a three-dimensional holographic model, while the dynamic AR model refers to holographic models that have been made into the three-dimensional animation. Some annotation symbols are also included within these five categories. In practice, images, videos, and static and dynamic AR models do not replace text but rather deepen and concretize key process steps described in the

text. This ensures that authoritative process instructions are preserved while the effectiveness and usability of guidance are improved through intuitive visual displays. It should be noted that, due to limitations such as the field of view and size of the AR device display, using all five representation forms for assembly procedure is not always optimal. An excessive combination of representation forms may lead to information redundancy. Therefore, it is necessary to balance the relationship between cognitive ability and information load.

Forming an effective information chain is a prerequisite for operators to accurately complete assembly task [57]. The brain's processing of AR assembly information can be divided into three stages: information perceiving, understanding, and information transmission. When presenting diverse AR representation forms to an operator, the operator first obtains primary information from the process text. This involves “perceiving” and “understanding” the process content. “Perceiving” refers to the brain's active filtering and focusing of key information based on existing cognitive patterns and expectations. It can be understood as the brain capturing the core elements of the assembly process text. For example, when viewing the LRM connector process text shown in Fig. 3, the perceived assembly objects include the LRM connector, printed board, pin, bolt set, etc., while the assembly actions include ‘place’, ‘align’, and ‘insert’, and the assembly path is ‘from the inside out of the rack’. “Understanding” is the process of deeply exploring and constructing various complex relationships between core elements based on perception, forming a complex network of connections. For example, in the sentence ‘place the printed board on the base of the workbench to measure that it will not move’ there is a close relationship between three core elements: the assembly action, the assembly object, and the assembly location. The assembly action places the object in a specific position to ensure its stability. Next, operators perform tasks in the real environment based on their perception and understanding. During this process, the information provided by AR representation forms aims to further help operators establish a connection with the real environment, which is called “information transmission”. However, when multiple types of representations are presented together, it is crucial to assess whether the core elements are clearly expressed and to what extent they are conveyed effectively.

The information load generated during the three processes of “perceiving”, “understanding”, and “transmission” can be represented separately as: assembly process perception load H_p , assembly process understanding load H_u , and assembly process information transmission load H_t . The first two terms are determined by the assembly process itself, while the last term is influenced by the AR representation forms used during the current assembly procedure.

Fig. 4 illustrates how the assembly process information load at the current assembly procedure is comprehensively quantified through the information perception load, understanding load, and transmission loads. The detailed definition, evaluation and calculation processes of each load are elaborated as follows.

4.2.1. Assembly process perception load

To effectively understand the assembly process information, operators first achieve an initial perception of the core elements in the assembly process. The types and quantities of core elements vary, and the frequency of each core element reflects its distribution, as well as the complexity of the assembly process text. This complexity

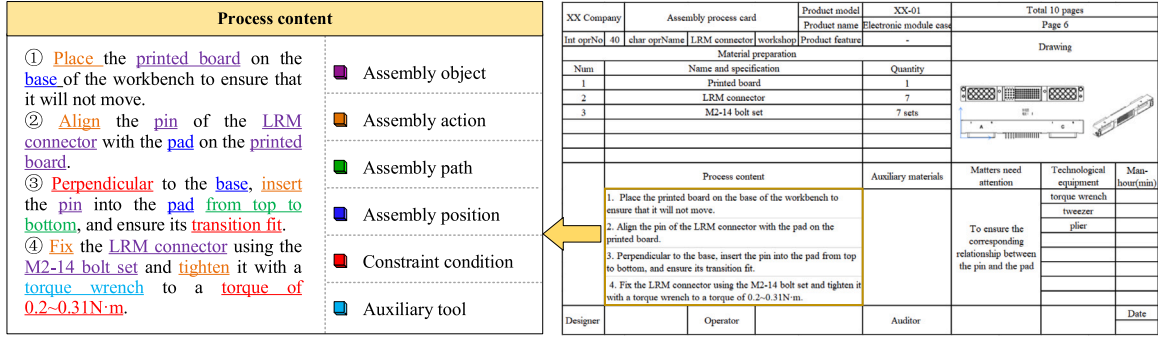


Fig. 3. Analysis of typical assembly process information.

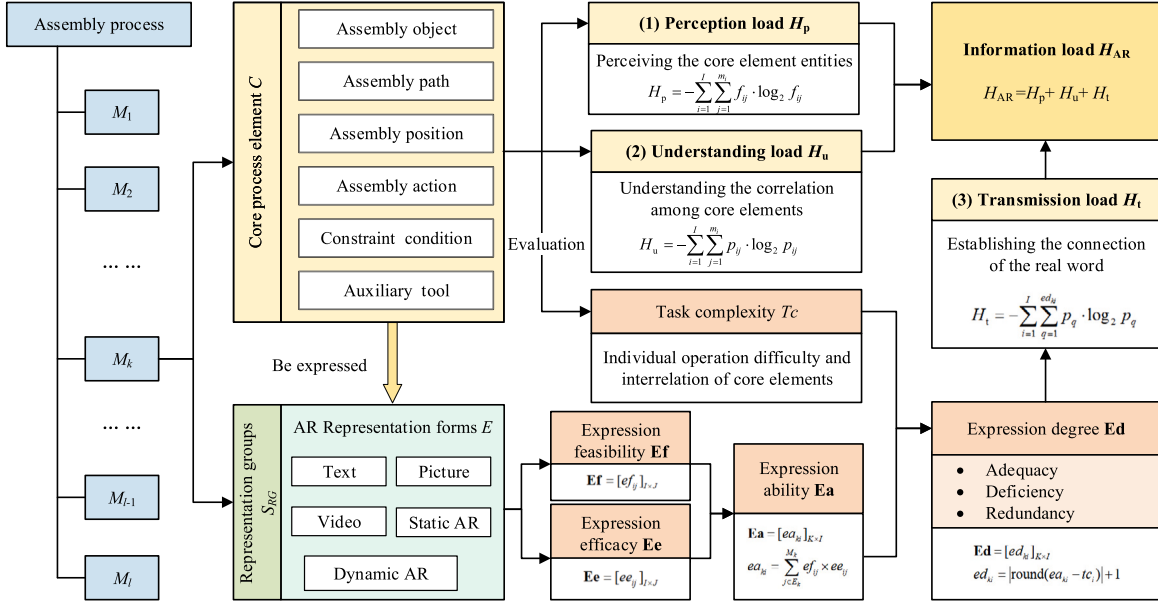


Fig. 4. Evaluation of assembly process information load.

introduces a cognitive load in the perception process, which we refer to as “perception load” and measure using information entropy.

Typically, an assembly operation involves some or all of six core elements, with different categories of entities within each core element type. Assuming that in procedure M_k , the i th core element type c_i has m_i categories of entities (i.e., there are m_i variable options for that core element type), and that the frequency of occurrence of core element entity c_{ij} is $f_{ij} = n_{ij}/N$, where n_{ij} is the number of occurrences of c_{ij} and N is the total number of occurrences of all core element entities, $N = \sum_{i=1}^I \sum_{j=1}^{m_i} n_{ij}$. The perception load for operation M_k can then be derived using Eq. (6):

$$H_p = - \sum_{i=1}^I \sum_{j=1}^{m_i} f_{ij} \cdot \log_2 f_{ij} \quad (6)$$

To illustrate the calculation process in detail, we take the assembly of the LRM connector shown in Fig. 3 as an example. This procedure involves four types of assembly object entities: the LRM connector, printed board, pin, and bolt set (i.e., $m_i = 4$). There are a total of 21 instances of core element entities in the entire procedure, with the LRM connector and printed board each appearing with a frequency of 2/21. The remaining core element entities each appear only once, with a frequency of 1/21. Using Eq. (6), the perception load for this procedure is calculated to be 3.916.

4.2.2. Assembly process understanding load

After perceiving the initial assembly information, the brain further processes and integrates this data to build a coherent understanding of the entire assembly process. Essentially, a worker's understanding of assembly process information involves grasping the relationships between core elements, which together create a network of interconnections. The proportion of relationships each core element entity has with other entities within the total set of relationships reflects the distribution of core element relationships, indicating the complexity of information during the comprehension process. We use information entropy to quantify the complexity of these relationships. Each core element contains multiple categories of entities, and relationships exist not only between entities within each core element but also between entities across different core elements. In other words, for each type of entity, we need to understand its connection to all other entities. As a result, the relationships between all core element entities, including those within the same element, constitute a relational matrix as shown in the Table 4. Specifically, the relationship $R(c_{ij}, c_{kl})$ between the j th entity c_{ij} of the i th core element and the l th entity c_{kl} of the k th core element is defined as Eq. (7):

$$R(c_{ij}, c_{kl}) = \begin{cases} 0 & c_{ij} \text{ and } c_{kl} \text{ have no relation} \\ 1 & c_{ij} \text{ and } c_{kl} \text{ have a relation} \end{cases} \quad (7)$$

Let $L_{ij} = \sum_{k=1}^I \sum_{l=1}^{m_k} R(c_{ij}, c_{kl})$, representing the sum of relationships between the core element entity c_{ij} and all other core element entities

Table 4
Calculation of understanding load of core elements.

$R(c_{ij}, c_{kl})$	c_{11}	c_{12}	c_{13}	c_{14}	c_{21}	c_{22}	c_{23}	c_{24}	c_{25}	c_{31}	c_{41}	c_{42}	c_{51}	c_{52}	c_{53}	c_{61}	L_{ij}
c_{11}	1	1	1	1	1	1	1	0	1	0	1	1	0	1	0	0	11
c_{12}	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	15
c_{13}	1	1	1	0	0	1	1	0	0	1	0	1	1	1	0	0	9
c_{14}	1	1	0	1	0	0	0	1	1	0	0	0	0	0	1	1	7
c_{21}	1	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	3
c_{22}	1	1	1	0	0	1	0	0	0	0	0	1	0	0	0	0	5
c_{23}	1	1	1	0	0	0	1	0	0	1	0	1	1	1	0	0	8
c_{24}	0	1	0	1	0	0	0	1	1	0	0	0	0	0	0	0	4
c_{25}	1	1	0	1	0	0	0	1	1	0	0	0	0	0	1	1	7
c_{31}	0	1	1	0	0	0	1	0	0	1	0	1	1	0	0	0	6
c_{41}	1	1	0	0	1	0	0	0	0	0	1	0	1	0	0	0	5
c_{42}	1	1	1	0	0	1	1	0	0	1	0	1	1	1	0	0	9
c_{51}	0	1	1	0	0	0	1	0	0	1	1	1	1	0	0	0	7
c_{52}	1	1	1	0	0	0	1	0	0	0	0	1	0	1	0	0	6
c_{53}	0	1	0	1	0	0	0	0	1	0	0	0	0	0	1	1	5
c_{61}	0	1	0	1	0	0	0	0	1	0	0	0	0	0	1	1	5
L_{ij}	11	15	9	7	3	5	8	4	7	6	5	9	7	6	5	5	112
p_i	0.101	0.138	0.083	0.064	0.028	0.046	0.073	0.037	0.064	0.055	0.046	0.083	0.064	0.055	0.046	0.046	1
H_u	3.956																

Note. c_{11} , c_{12} , c_{13} , and c_{14} represent the assembly objects: printed board, LRM connector, pin, and bolt set, respectively. c_{21} , c_{22} , c_{23} , c_{24} , and c_{25} denote the assembly actions: place, align, insert, fix, and tighten. c_{31} indicates the assembly path: from top to bottom. c_{41} and c_{42} specify the assembly locations: base and pad. c_{51} , c_{52} , and c_{53} represent constraint conditions: perpendicular, transition fit, and torque. Finally, c_{61} refers to the auxiliary tool: torque wrench.

(including itself). Therefore, $L = \sum_{i=1}^I \sum_{j=1}^{m_i} L_{ij}$ represents the total sum of relationships among all core element entities. Based on the information entropy formula, the cognitive load generated by the operator during the understanding of process information is expressed in Eq. (8). Here $p_{ij} = L_{ij}/L$ denotes the probability of understanding the relationship between a core element entity c_{ij} and other core element entities relative to all relationships. Take the process depicted in Fig. 3 as an example. Table 4 illustrates the relationships among the core process elements. Based on the Eq. (8), the final calculation shows that the cognitive load for understanding the information in this process is 3.956.

$$H_u = - \sum_{i=1}^I \sum_{j=1}^{m_i} p_{ij} \cdot \log_2 p_{ij} \quad (8)$$

4.2.3. Assembly process information transmission load

This load is generated during the association of AR representation forms with the assembly task, and is measured using information entropy. When assembly process information is conveyed through different AR representation forms, the brain needs to match this information with the actual assembly tasks to establish an effective operational process. The load depends on whether the combination of AR representation forms is suitable for the assembly tasks—whether it results in information shortage, adequacy, or redundancy. This reflects the level of uncertainty that the brain faces in matching the information.

To be specific, in terms of AR representation forms, different representations vary in their ability to convey different core elements. If a representation form can effectively convey core elements and make them easily understandable, its expression ability is considered strong. In terms of the complexity of the core elements being expressed, if the combination of representation forms adequately matches the complexity of the core element, the combination is considered appropriate. Otherwise, it may be redundant or deficient, leading to multiple interpretations during the transmission of information to the brain, thereby generating cognitive load. Therefore, the calculation of this load includes three stages:

(a) Quantifying the expression ability

Based on Eq. (5), the combination of representation forms can be abbreviated as representation group. The representation group is the

combination of “text” along with other four representation forms, resulting in a total of 15 combinations. The set of these 15 representation groups is denoted as Eq. (9):

$$S_{RG} = \{RG_1, RG_2, \dots, RG_k, \dots, RG_K\} \quad (9)$$

where $K = 15$. For the k th representation group, $RG_k = \{e_{k,n} | n = 1, 2, \dots, N_k\}$, where $e_{k,n}$ represents each type of representation form such as text, images, etc., and N_k represents the number of representation forms included in RG_k . When $e_{k,n} = e_j$, these representation forms are indexed as j in the set E , and the set of these indices can be denoted as $E_k = \{j | e_{k,n} = e_j\}$.

The expression ability of representation group is jointly decided by both the expression feasibility and the expression efficacy of the representation forms within the group. Therefore, we define the following three terms to quantify the expression ability of representation group.

- **Expression feasibility (Ef):** Whether a representation form e_j can clearly express the core element c_i .
- **Expression efficacy (Ee):** The degree to which a representation form e_j can clearly express the core element c_i .
- **Expression ability (Ea):** A combination of expression feasibility and expression efficacy, measuring the extent to which representation group RG_k can effectively and clearly expresses core elements.

Based on the dimensions and states of representation forms, both representation forms and core elements can be categorized into four types: static 2D, dynamic 2D, static 3D, and dynamic 3D. For representation forms, text and images belong to static 2D, videos to dynamic 2D, static AR models to static 3D, and dynamic AR models to dynamic 3D. For different categories of core elements, these four types represent the simplest form that can achieve clear expression of core elements. Through extensive analysis of assembly process documentation and insights from workers' years of practical experience, we identified a pattern in the representation of core elements across different dimensions and states. For most assembly objects and auxiliary tools, static 2D representations provide sufficient information. However, for more complex core elements such as assembly locations and constraints, higher dimensional and dynamic representations are often required for clarity. In the context of electronic equipment assembly, the simplest form for clearly conveying assembly objects (e.g., connectors, printed

Table 5
Different representation forms for the state of core elements Ef matrix.

Expression feasibility	Static 2D		Dynamic 2D		Static 3D	Dynamic 3D
	Assembly object	Auxiliary tool	Assembly action	Assembly path	Assembly position	Constraint condition
Text (Static 2D)	1	1	1	1	1	1
Picture (Static 2D)	1	0	0	0	1	0
Video (Dynamic 2D)	1	1	1	1	1	0
Static AR model (Static 3D)	1	0	0	0	1	1
Dynamic AR model (Dynamic 3D)	1	1	1	1	1	1

boards) and auxiliary tools (e.g., torque wrenches, measuring instruments is Static 2D; for assembly actions and paths, it is Dynamic 2D; for assembly locations (e.g., Grooves and holes), it is Static 3D; and for constraint conditions, it is Dynamic 3D. According to the order “static 2D, dynamic 2D, static 3D, and dynamic 3D”, the clarity of representing core elements through each form increases. Conversely, for core elements themselves, the difficulty of being clearly expressed also increases in this sequence.

A sample Ef matrix (Eq. (10)) is shown in Table 5, demonstrating the feasibility to express elements in a given scenario. Note that this table exemplifies a specific context, and Ef can fluctuate depending on individual comprehension and the varying states of product expression. Therefore, the values in the matrix should be adjusted according to real-world circumstances.

$$\mathbf{Ef} = [ef_{ij}]_{I \times J} \quad (10)$$

$$ef_{ij} = \begin{cases} 0 & e_j \text{ cannot express the core element } c_i \\ 1 & e_j \text{ can express the core element } c_i \end{cases} \quad (11)$$

where, ef_{ij} represents the expression feasibility of representation form e_j to the core process element c_i .

The higher the probability that a core element is clearly conveyed, the higher the expression efficacy of the representation form for that element. We define the state of representation form e_j (static or dynamic) as Se , and its dimension (2D or 3D) as De . Similarly, the state of core element c_i is defined as Sc and its dimension as Dc . As the values of Se and De increase, the probability q_{ij} that representation form e_j can clearly express core element c_i also increases, so the more certain the event “the expression form clearly expresses the core element” becomes. Conversely, as the values of Sc and Dc increase, the difficulty in clearly expressing core element c_i increases, and the probability q_{ij} decreases, thereby increasing the uncertainty of the event, which in turn increases the information content of the event. The information content contained in this event is given by [58]:

$$Ia_{ij} = -\log_2 q_{ij} \quad (12)$$

where, $q_{ij} = (Se + \alpha De) / (Sc + \alpha Dc)$ represents the probability that representation form e_j can express core element c_i . It should be noted that after calculating all the q_{ij} values, normalization needs to be performed. α is used to distinguish the importance degree of dimension and state. In this paper, we consider the impact of 3D versus 2D to be greater than that of dynamic versus static, so α is set to be a constant less than 1, $\alpha = 0.8$. The higher the information content, the greater the uncertainty of the event, and the lower the expression efficiency. To quantify the expression efficacy of representation forms for core elements, the Expression efficacy matrix (Ee) is derived as Eq. (13), where, ee_{ij} represents the expression efficacy of the representation form e_j on the core element c_i .

$$\mathbf{Ee} = [ee_{ij}]_{I \times J}, ee_{ij} = \frac{1}{Ia_{ij}} \quad (13)$$

Finally, the expression ability matrix (Ea) for representation group RG_k express the core element c_i is calculated by Eq. (11), where, Ea is derived by element-wise multiplication of matrices Ef (Eq. (10)) and Ee (Eq. (13)).

$$\mathbf{Ea} = [ea_{ki}]_{K \times I}, ea_{ki} = \sum_{j \in E_k}^{N_k} ef_{ij} \times ee_{ij} \quad (14)$$

(b) Calculating the expressive intensity

The above calculation of expression ability applies to general situations. When addressing specific tasks, for procedure M_k , the clarity of its expression using representation forms can fall into three categories: (a) Information deficiency, where the information is insufficient or too brief to explain the assembly process, making it difficult to be comprehended. (b) Information adequacy, where the information accurately describes the main event without redundancy or deficiency. (c) Information redundancy, which, despite explaining the main event, increases the uncertainty in understanding it. The function of representation forms is to help operators understand the assembly process to perform the correct operations. Whether the representation is deficient, adequate, or redundant should be evaluated based on the complexity of the current assembly task.

The task complexity of a procedure refers to the comprehensive assessment of the interrelationships among the core elements and the difficulty of individual operations for the procedure. This evaluation encompasses multiple factors, including the relative positions of assembly object, the assembly path, the technical requirements of assembly action, the accessibility of assembly positions, the types and precision of constraints, and the operational difficulty of auxiliary tools.

By comprehensively considering the independent completion difficulty and the interactions among these elements, a thorough assessment of the complexity of the assembly task can be achieved. As shown in Table 6, this assessment breaks down the contributing factors into graded scales and quantifiable criteria for each element, enabling a systematic evaluation of operational complexity and guiding enhancements to assembly processes. The task complexity of a specific operation can thus be represented as:

$$Tc = \sum_{i=1}^I tc_i \quad (15)$$

where, $tc_i = \sum_{d=1}^D (r_d \cdot w_d)$ refers to the complexity of a single core element. For each core element c_i , its complexity is determined by the product of the ratings r_d and weights w_d across D dimensions.

In summary, the expression intensity $ed_{k,i}$ is defined as the degree to which representation group RG_k can clearly express the core element c_i when considering task complexity. Therefore, the Expression intensity matrix (Ed) is given by Eq. (16), where round ($\cdot$) denotes rounding to the nearest integer.

$$\mathbf{Ed} = [ed_{ki}]_{K \times I}, ed_{ki} = \lceil \text{round}(ea_{ki} - tc_i) \rceil + 1 \quad (16)$$

When round ($ea_{ki} - tc_i$) = 0, it indicates that the information contained in RG_k is just sufficient to clearly express c_i given its complexity, corresponding to “information adequacy” state. round ($ea_{ki} - tc_i$) < 0 indicates that RG_k lacks the necessary information to express c_i clearly, corresponding to “information deficiency” state. Conversely, when round ($ea_{ki} - tc_i$) > 0, it shows that RG_k contains more information than needed to express c_i given its complexity, potentially causing interpretative interference, indicating an information redundancy state. Here, the constant 1 is an additional constant for expression intensity, indicating a situation where RG_k can clearly express c_i . From the operator’s perspective, it refers to a situation where the task content described by the core element c_i is clearly understood. If $ed_{k,i}$

Table 6
Task complexity rating.

Core elements	Metrics	Weight	Rating: 1–5				
			1	2	3	4	5
Assembly object	Types and quantity	0.5	Single	Few	Moderate	Multiple	Heterogeneous
	Size	0.25	Small	Medium	–	Large	Very large
	Shape	0.25	Simple	–	Complex	Irregular	Special shapes
Assembly path	Path direction	0.5	Single direction	Simple direction change	Moderate change	Complex direction change	Highly complex
	Path obstacles	0.5	No Obstacles	Few obstacles	Moderate obstacles	Many obstacles	Very many obstacles
Assembly action	Skill requirements	0.2	Basic skills	Simple skills	Moderate skills	Professional skills	Highly specialized skills
	Action complexity	0.3	Simple actions	Coordinated actions	Moderate actions	Complex actions	Highly coordinated
	Number of actions	0.5	Few	Moderate	Many	Numerous	Very numerous
Assembly position	Accessibility	1.0	No restriction	Simple restriction	Moderate restriction	Complex restriction	Highly restrictive
Constraint condition	Moment constraints	0.3	Fastener preload	Assembly torque	Preload constraints	Multi-point torque	Stress distribution
	Environmental constraints	0.3	Temperature	Humidity	Cleanliness	Air pressure	Vibration
	Tolerance constraints	0.4	Position tolerance	Symmetry tolerance	Shape tolerance	Fit tolerance	Parallelism or perpendicularity
Auxiliary tools	Tool complexity	0.5	No tools	Simple tools	Intermediate tools	Complex tools	Precision tools
	Usage requirements	0.5	Easy operation	Moderate operation	Complex operation	–	Professional technique

Table 7
The calculation process of information transmission load.

Core process elements	Static 2D		Dynamic 2D		Static 3D	Dynamic 3D
	Assembly object	Auxiliary tool	Assembly action	Assembly path	Assembly position	Constraint condition
$ea_{k,i}$	2.94	1.74	1.15	1.15	1.91	1.46
tc_i	2.05	2.5	1	2.1	2.55	1
$ed_{k,i}$	2	2	1	2	2	1
p_q	0.50	0.50	1.00	0.50	0.50	1.00
H_t	4					

$= 2 + 1$, it implies that, besides the clear expression of the core element, there are two other scenarios that may lead to different interpretations or misunderstandings (misunderstandings is also a form of understanding).

(c) Calculating the transmission load

Based on the above definitions, for the core process element c_i within the process information, there are $ed_{k,i}$ possible ways that an operator might understand it. Assuming the probability of understanding each possible interpretation is equal, the probability that an operator can correctly understand the conveyed process information is given by:

$$p_q = \frac{1}{ed_{k,i}} \quad (17)$$

Therefore, the information transmission load, which represents the uncertainty generated by the representation group in clearly conveying the core element considering task complexity, is given by:

$$H_t = - \sum_{i=1}^I \sum_{q=1}^{ed_{k,i}} p_q \cdot \log_2 p_q \quad (18)$$

To facilitate understanding, consider the example in Fig. 3 of the procedure using the representation group “text + video + static AR model”. The calculation of transmission load is shown in Table 7.

In summary, after calculating the information perception load H_p , understanding load H_u , and transmission load H_t , the total information load H_{AR} for a given assembly procedure M_k and AR representation group RG_k is:

$$H_{AR} = H_p + H_u + H_t \quad (19)$$

4.3. Inference of assembly state based on context awareness

Assembly state denotes the stages within an assembly procedure, such as start, in progress, and stop. A context-aware approach is proposed for inferring the “assembly state” automatically such that the information can be pushed proactively and implicitly without excessive interaction by operators. Context-awareness specifically focuses on the recognizing the operator’s actions and detecting the types of parts, ensuring a comprehensive understanding of the assembly task. In this paper, the assembly state focuses on the state of assembly in progress and stop, represented by M_k -keep and M_k -stop respectively.

Each procedure can be characterized including its unique actions and parts. However, solely relying on action recognition or part detection may not ensure accurate judgment, as some procedures might involve similar actions, such as the “tightening” action required for installing fasteners. And adjacent procedures might use the same parts, e.g., “cable connectors” are used in both procedures of cable assembly and connector assembly. This overlap necessitates a joint analysis of actions and objects for accurate procedure determination. Therefore, to identify the assembly procedure, action recognition and part detection algorithms are jointly employed.

act_i and $part_j$ respectively represent the subsets of the sets Act , and $Part$, which denote assembly actions and parts. And the M_k represent the k th assembly procedure of the assembly process M :

$$\begin{cases} Act = \{act_1, act_2, \dots, act_i, \dots, act_m\} \\ Part = \{part_1, part_2, \dots, part_j, \dots, part_n\} \\ M = \{M_1, M_2, \dots, M_k, \dots, M_l\} \end{cases} \quad (20)$$

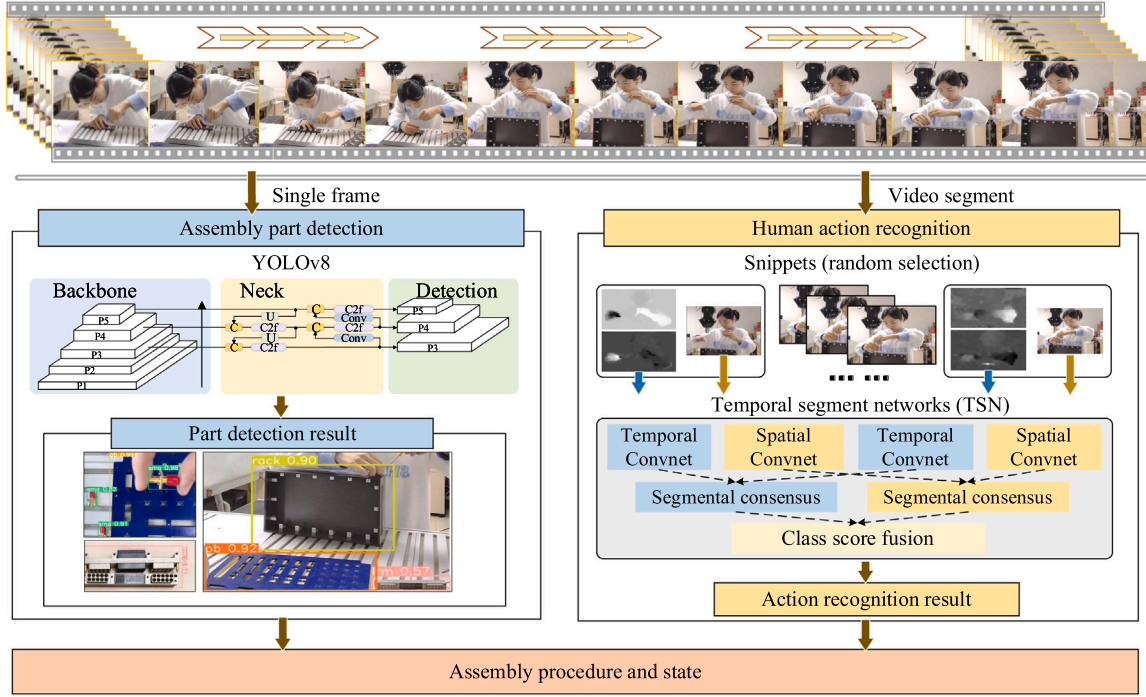


Fig. 5. Assembly state inference based on context awareness.

The corresponding procedure M_k is determined by using the detected assembly action (act_i) and part ($part_j$). In addition, to further enhance system reliability and accuracy, we introduced Human-computer Interaction (HCI) mechanism with buttons, voice commands, and gaze functionality, which allows operators to manually confirm or correct detection results. This leverages the operators' expertise and experience, compensating for potential shortcomings in automated detection, and ensuring more reliable assembly guidance.

Using action recognition and part detection for joint inference of assembly states is illustrated in Fig. 5. This study employs a Temporal Segment Network (TSN) based on transfer learning for action recognition [59]. TSN is based on the idea of long-range temporal structure modeling. It combines a sparse temporal sampling strategy and video-level supervision to enable efficient and effective learning using the whole action video. The manual assembly process is time-consuming with low actions frequency, i.e., the actions do not change frequently. The core advantage of TSN lies in their ability to effectively model long-term temporal dynamics, efficiently processing video frames using pre-trained image classification networks. This significantly enhances video action recognition performance across multiple datasets without requiring a large amount of labeled data.

To enhance the performance of the TSN algorithm, we applied a pre-trained model from the Kinetics-400 dataset for transfer learning, leveraging its extensive video coverage to enhance feature representation and mitigate overfitting. This strategy improves generalization and particularly valuable when training data is limited.

The state-of-the-art (SOTA) part detection model YOLOv8 is trained to for detecting parts types. YOLOv8 employs the C2f structure (CSPLayer_2Conv), which effectively reduces computational load and improves convergence speed and efficiency. YOLOv8's superior speed and accuracy in part detection ensure reliable identification of parts during assembly.

The process of online video data acquisition, segmenting, sensing, joint assembly state reasoning is detailed in Algorithm 1. Due to the need for detection over a period, different functions are executed by

multiple threads to ensure the safe and sequential handling of the queue data.

Thread 1 is responsible for processing video frames. The object detection task focuses on each frame of the video stream, while the action recognition task requires a sequence of video frames over a period. Supposing the video stream V has a frame rate of n fps (frames per second), with the video stream arriving, the video frame is added to the detection queue R as well as to the list L_t . Let T denotes the duration of the video sequence for action recognition, thus the number of frames is $T \times n$. To ensure continuity of frames during action recognition, L_t needs to incorporate half of the preceding video sequence to form the new video sequence. Therefore, L_t is calculated as Eq. (21):

$$L_t = \left\{ L_{t-1} \left(\frac{T \times n}{2} + 1 \right), \dots, L_{t-1}(T \times n), f_{\text{start}}, \dots, f_i, \dots, f_{\text{end}} \right\} \quad (21)$$

where, $L_{t-1}(T \times n/2 + 1)$ to $L_{t-1}(T \times n)$ represents the latter half of the previous video sequence L_{t-1} , while f_{start} to f_{end} denotes the starting and ending frames constituting the new sequence, ensuring their continuity. After generating the list, place it into the action recognition queue P .

Thread 2 is used to obtain results from object detection. It takes the video frame f_i from queue R and passes it through the preprocessing pipeline. The YOLOv8 model is used for object detection, recording all detection results within $T \times n$ timestamps. The part detection results from YOLOv8 are stored in a queue named TC .

Thread 3 is used to retrieving results from action recognition. It takes the video frame list L_t from queue P and feeds it into the preprocessing pipeline. The TSN model is used for action recognition, and the action recognition results from TSN are stored into queue AC .

Thread 4 handles joint inference results. It retrieves aggregated results from queues TC and AC , performs joint inference, and obtains the final inference result.

Algorithm 1 Online Assembly Process Inference.

Input: Live video stream (V), Video frame rate (n), Pre-processing pipeline (S), Action recognition model (TSN), Object detection model (YOLOv8)

Output: Assembly process predictions (M_k)

1: Initialize video queue Q for n incoming frames; Initialize list L_i for action recognition; Initialize queue P for action recognition (r video segments); Initialize queue AC for action recognition (r video segments); Initialize queue R for target detection; Initialize queue TC for target detection

2: **THREAD_1: Get Live Video Stream**

3: **while** new frames available from V **do**

4: Add frame q_i from V to queue R

5: $t = \text{int}(i/n)$

6: Add frame q_i from V to queue L_i

7: **if** $t == 0$ **then**

8: $L_i = L_0$

9: **else**

10: $L_i = \left\{ L_{i-1} \left(\frac{T \times n}{2} + 1 \right), \dots, L_{i-1}(T \times n), f_{\text{start}}, \dots, f_i, \dots, f_{\text{end}} \right\}$

11: **end if**

12: **if** $\text{len}(L_i) == T \times n$ **then**

13: Add frame L_i to queue P

14: **end if**

15: **end while**

16: **THREAD_2: Get Target Detection**

17: **while** frames available from R **do**

18: Feed frame r to S and YOLOv8 to obtain prediction

19: Add the target detection result to TC

20: **end while**

21: **THREAD_3: Get Action Recognition Result**

22: **while** framelist available from P **do**

23: Feed framelist r to S and TSN to obtain prediction

24: Add the action recognition result to AC

25: **end while**

26: **THREAD_4: Get Joint Reference**

27: **while** action recognition available from AC and target detection available from TC **do**

28: $act_i = AC.\text{Dequeue}()$

29: $part_j = TC.\text{Dequeue}()$

30: Combine action prediction and object detection results to obtain assembly process state M_k

31: **end while**

4.4. Alignment of information load to cognitive ability

To achieve adaptive information push in assembly tasks, this section develops an adaptive information push mechanism based on the Cognitive Load Theory (CLT) [60] and the design principles of Adaptive User Interfaces (AUI) [61]. This mechanism dynamically adjusts AR content to align with the user's cognitive abilities, enabling implicit interaction (i.e., real-time perception of the assembly states and triggering of information push) and optimizing information presentation to reduce the operator's cognitive load. For AR content to meet these objectives, four conditions must be satisfied:

(1) A specific assembly procedure M_k , ensuring that the AR content displayed aligns with the current procedure in the assembly process.

(2) A specific cognitive ability C_A , based on the user's skill level and fatigue, to tailor content to the operator's needs.

(3) The expression ability Ea_k of each representation group, corresponding to the user's skill level in C_A , users with lower skill levels benefit more from high-expression content.

(4) The cognitive load H_{AR} of each representation group, corresponding to the user's fatigue level, as users experiencing high fatigue benefit from low-load content.

By ranking the information loads for all the 15 representation groups, the information load levels are established. On this basis, the

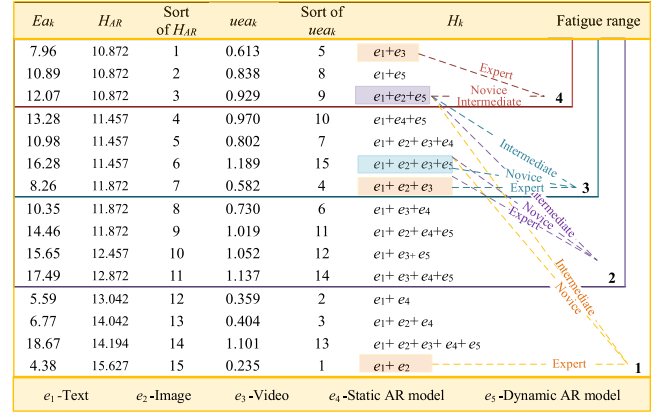


Fig. 6. The execution process of the information push mechanism.

optimal assembly information H_k of the current procedure M_k should be selected based on a comprehensive consideration of the operator's cognitive ability C_A , the expression ability (Ea) of the information, and the overall information load, as shown in Eq. (22):

$$H_k = g(M_k, C_A, Ea_k, H_{AR}) \quad (22)$$

where, C_A refers to the cognitive ability of the current user (influenced by skill and fatigue). Based on Eq. (14), is the expression ability of representation group RG_k express the procedure, H_{AR} refers to the information load of the current procedure M_k . For operators, without considering fatigue levels, stronger expression ability Ea_k generally aids in better understanding of process information. However, excessively strong or weak of expression ability increase cognitive load. Therefore, the determination process of H_k should follow the two principles: (1) Higher levels of fatigue incline towards selecting RG_k with lower information load, and vice versa. (2) Lower skill levels incline towards selecting RG_k with stronger expressive abilities, and vice versa. When considering the fatigue level, the information load H_{AR} levels are classified into four ranges. With the increase of fatigue level, the selection range when pushing information is narrowed down. When considering the skill level, the selection range when pushing information should be based on the expressive abilities of the information. We define Eq. (19) to describe the expression ability of unit information load. Note that the values of Ea_k and H_{AR} need to be normalized.

$$uea_k = \frac{Ea_k}{H_{AR}} \quad (23)$$

The higher the uea_k is, the easier the information is to understand, which means the representation groups are more diverse and thus is more suitable for novices. Conversely, the lower the uea_k is, the more difficult the information is to understand, indicating that the representation combination is simpler and more suitable for experts. A value of uea_k close to 1 indicates that the representation combination is "just right", neither deficient nor redundant, which is suitable for intermediate with some experience.

Taking the procedure Fig. 3 as an example, we detail the process of the information push mechanism as follows. The 15 types of representation groups are reported in the 6th column of Fig. 6. The expression ability values Ea_k of each representation group, the calculated information load values H_{AR} , the expression ability of unit information load uea_k , and the uea_k rankings are also presented in Fig. 6. The fatigue level is considered as a constrain for information push. When the fatigue level of the user is 1, the acceptable range of information for this user includes all these representation group. For experts, simpler information is better, so the combination with the lowest uea_k ranking, " $e_1 + e_2$ " will be pushed to them. For intermediate, the combination

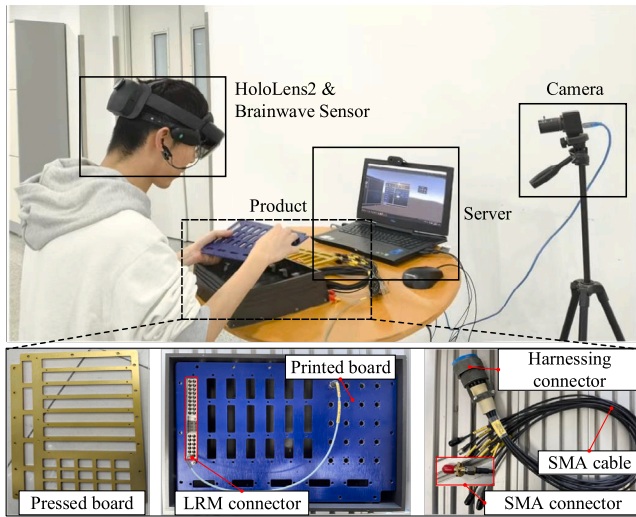


Fig. 7. Experimental setup and parts.

Table 8
Parts and their identifiers.

Part	Identifier
Bracket	Part1
Side panel	Part2
Crate	Part3
Printed board	Part4
Pressure board	Part5
LRM connector	Part6
SMA connectors	Part7
Cables	Part8

uea_k with closest to 1, “ $e_1 + e_2 + e_5$ ” will be pushed. For novices, the combination with the strongest expressiveness, “ $e_1 + e_2 + e_3 + e_5$ ” is preferred. When the fatigue level is 2, the push principle remains the same, based on the range provided in the table. Then the representation group with lowest uea_k i.e., “ $e_1 + e_2 + e_3$ ”, will be pushed to experts. The one with the uea_k value closest to 1 will be pushed to intermediate, the one with highest uea_k value will be pushed to novices. Similarly, when the fatigue level is 3, “ $e_1 + e_2 + e_3$ ” will be pushed to experts, “ $e_1 + e_2 + e_5$ ” to intermediate, and “ $e_1 + e_2 + e_3 + e_5$ ” to novices. When the fatigue level is 4, “ $e_1 + e_3$ ” will be pushed to experts, “ $e_1 + e_2 + e_5$ ” to intermediate, and “ $e_1 + e_2 + e_5$ ” to novices.

5. Experiments and user study

To demonstrate the feasibility and effectiveness of our proposed method, we conducted experiments on an AR-guided electronic equipment assembly task. As illustrated in Fig. 7, the system comprises a USB camera, HoloLens 2, Sichiray Brainwave Sensor, CPU server (Intel i7-10700K), and GPU server (NVIDIA GeForce RTX 3060 Ti, 8 GB). The system was developed in a Windows 10 (64-bit) environment. The TSN and YOLOv8 models were implemented using Python 3.9 within the PyTorch framework. The AR application was developed with the Unity 3D. The assembly task involved parts such as brackets, side panels, crate, printed board, pressure board, LRM connectors, SMA connectors, and their respective cables, as shown in Table 8. The relationship between each assembly procedure and the involved parts is shown in Table 9.

5.1. Fatigue state perception experiment

To accurately monitor the operator’s fatigue state in real-time, an experiment of continuous cognitive task training is conducted. A total

Table 9

The corresponding relationship of procedures, parts, and actions.

Assembly procedures	Corresponding parts	Iconic actions
M1-Side panel assembly	Part1, Part2	Act1 - Screw bolts assembly
M2-Crate assembly	Part3	Act2 - LRM connector assembly
M3-Connectors assembly	Part4, Part6, Part7	Act3 - SMA connector assembly
M4-Board assembly	Part3, Part4, Part5	Act4 - Board assembly
M5-Cable assembly	Part3, Part5, Part8	Act5 - Cable assembly

Table 10

Classification accuracy of different models.

Classifier	Random Forest (RF)	Support Vector Machine (SVM)	Logistic regression	Artificial neural network (ANN)
Accuracy (%)	87.7	81.4	79.9	81.9
Precision (%)	86.5	80.2	77.9	80.5
Recall (%)	82.5	78.6	76.3	79.2
F1-Score (%)	84.1	79.4	76.9	79.8

of 12 healthy volunteers, aged between 20 and 35, were selected, including undergraduate, master’s, and doctoral students. These participants were chosen for their strong learning and comprehension abilities, which allowed them to follow the experimental procedures effectively. Additionally, individuals in this age range tend to have stable physical and cognitive capabilities, which contributes to the reliability and accuracy of the experimental data. The participants came from diverse fields and had varying levels of familiarity with and skill in assembly-related cognitive tasks. However, since the focus of this study is on collecting brainwave data during cognitive tasks, these individual differences did not impact the fatigue analysis.

Before the experiment, participants were fully informed about the experimental content, procedures and objectives in advance. They kept working continuously from 8 AM to 11 PM. Every three hours they were required to fill out a fatigue questionnaire and a 6-min brainwave signal were acquired. Only the stable 5 min in the middle part of brainwave signals were used for analysis. For each fatigue state, 32 data segments were collected, each consisting of 5-min brainwave signals segmented at 2-s intervals. This resulted in 4800 data points per fatigue state. Four classification models are tested and the results are reported in Table 10. According to the results, the Random Forest (RF) is selected.

5.2. Accuracy of joint inference of assembly states

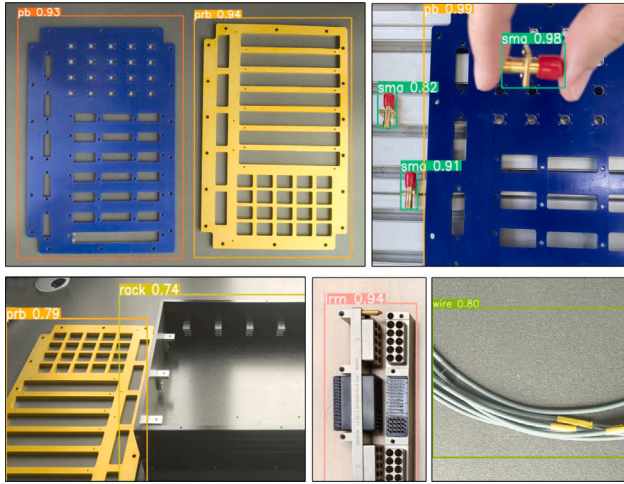
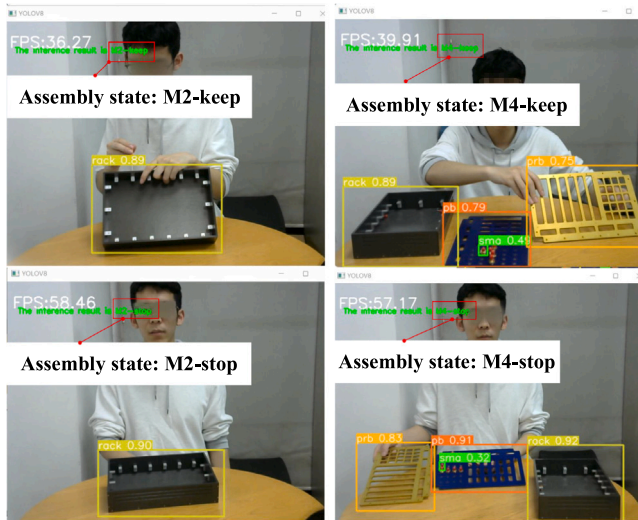
The action recognition model TSN and the object detection model YOLOv8 were tested to verify their abilities to accurately recognize the operator’s actions and to detect assembly parts. Action recognition videos of the operator’s assembly actions were collected from various perspectives, including front, left-front, and right-front angles. These perspectives ensured that the relevant parts were within view. This video dataset, named ARDT, was used for action recognition. The performance of the TSN model was evaluated using the Top-1 accuracy metric. The dataset comprised 1008 video segments, collected from 6 operators, with a resolution of 1280×720 at a frame rate of 30 fps, and each segment lasting 8 s. For each action type, 3/4 of the videos were used for the training dataset and 1/4 for the validation set. Additionally, 284 video segments were collected as test dataset, with average Top-1 accuracy 89.12%. As shown in Table 11.

To evaluate the YOLOv8, we sampled frames from the action recognition videos. A total of 760 images containing six types of parts were extracted. The dataset was further augmented and expanded using techniques such as scaling, flipping, center cropping, brightness adjustment, contrast adjustment, and noise addition. This resulted in an expanded dataset of 6579 images, designated as PDDT. From this dataset, 70% were randomly selected for training and 30% for testing. We used Average Precision (AP) and mean Average Precision (mAP) as

Table 11

The action recognition accuracy of the test dataset.

Action	Top-1	Sample size
Act1-Screw bolts assembly	98.28%	59
Act2-LRM Connectors assembly	66.54%	60
Act3-SMA Connectors assembly	100%	59
Act4-Board assembly	90.78%	56
Act5-Cable assembly	90%	50
Average	89.12%	284

**Fig. 8.** Results of object detection for some parts.**Fig. 9.** Some results of joint inference.

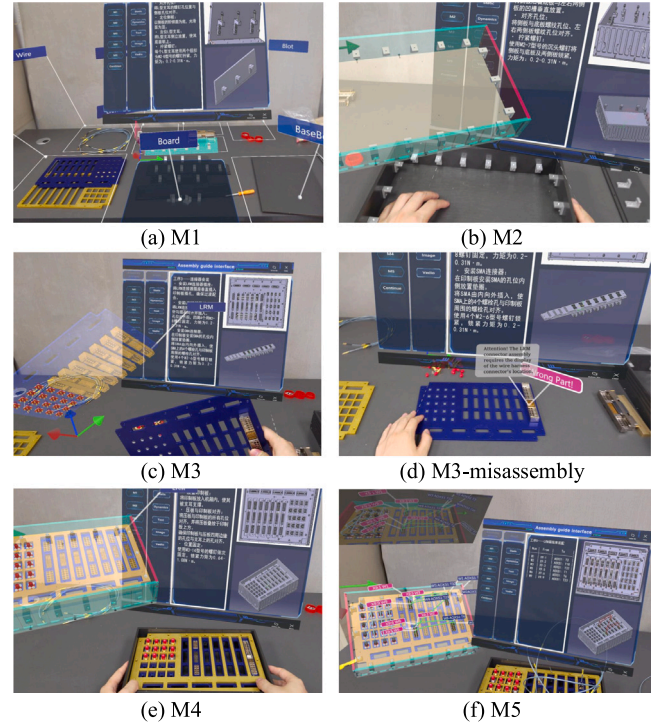
performance metrics for the model. The object detection results under different IoU (Intersection over Union) are shown in Table 12. Fig. 8 shows the detection results for various parts using YOLOv8.

Finally, joint inference experiment was conducted using the action recognition test dataset. The assembly state was inferred by both action recognition and part detection, specifically for the “keep” state of a procedure. When no actions were detected, the “stop” state was tested by checking the detected parts types. Fig. 9 illustrates the joint inference for procedures M2 and M4, with the inference results shown in green text. After testing, the average accuracy of the assembly state joint inference was 90.88%.

Table 12

Parts detection results.

Parts	Sample size	AP	mAP
Bracket	2592	93.40%	
Side panel	1809	99.40%	
Crate	999	98.30%	
Printed plate	936	99.40%	96.60% (IoU = 0.5)
Pressure plate	927	98.40%	72.80% (IoU = 0.5~0.95)
LRM connector	2961	94%	
SMA connector	2187	92.30%	
Cable	2606	97.40%	

**Fig. 10.** The process of AR-guided assembly operation.

5.3. Comprehensive evaluation of AR-AIP

5.3.1. Experimental design

To evaluate the effectiveness of the proposed method while identifying its advantages and limitations, we conducted a comparative experiment on two AR-guided assembly methods: AR-Base and AR-AIP. AR-Base is the conventional AR-guided assembly method using button clicks for interaction. The two object metrics, i.e., task completion time (TCT) and the number of errors (NoE), as well as two subject metrics, i.e., NASA-TLX [62] and System Usability Scale (SUS) questionnaires [63] are used to compare AR-Base and AR-AIP. Fig. 10 shows the five procedure of AR-guided assembly operation.

Previous studies [26] have shown that in the AR-Base control group, operators with different cognitive abilities exhibit variations in completion time, error rate, and cognitive load. Our method aims to reduce these differences by aligning with cognitive ability. We assume that cognitive ability is mainly determined by skill level and is constrained by fatigue level. AR-AIP takes both into account and adjusts the process information load accordingly. In the group experiments, fatigue levels and skill levels were considered independent variables. The following hypotheses are proposed. The experimental results will verify whether these hypotheses hold.

H1: The AR-AIP group will have a shorter TCT compared to the AR-Base group, reducing the TCT difference caused by skill level and fatigue.

H2: Considering fatigue and skill levels, AR-AIP will provide more accurate process information, reducing errors due to insufficient or overloaded information, and minimizing NoE differences caused by skill level and fatigue.

H3: The cognitive load in the AR-AIP group will be lower than in the AR-Base, and AR-AIP will reduce the cognitive load differences caused by skill level and fatigue.

H4: AR-AIP will offer a better system experience.

H5: AR-AIP will enhance operator satisfaction and work efficiency.

We recruited 48 volunteers from different majors and fields (ages 20–35) for the experiment, seeking participants with AR experience to minimize novelty effects. During recruitment, we collected information on participants' assembly experience and daily fatigue patterns through a questionnaire to facilitate grouping by fatigue and skill levels. Experience data included years of work, types of assembly tasks previously performed, and proficiency levels. According to the criteria, participants were divided into three groups: experts, who had extensive general assembly experience but were unfamiliar with this product type; intermediates, who had some relevant experience but were also new to this product; and novices, who had no prior assembly experience. Fatigue levels were determined based on the criteria outlined in Section 4.1, with participants grouped according to their self-reported fatigue levels. Before experiment, participants wore brainwave monitoring devices to confirm their current fatigue state and completed the MFI-20 scale, with their fatigue stability observed over a short period. For those who did not meet the target fatigue levels, we used mild cognitive activities to induce fatigue, ensuring data accuracy and consistency.

To avoid the learning effect, each participant conducted the experiment in an isolated space, unable to observe others' operations. The experiment was divided into 24 batches, with each batch including two participants of the same skill level to ensure consistency in experimental conditions. Fig. 12 shows scenes of three different types of operators using AR-AIP for assembly under severe fatigue. At this stage, M4 has been completed, and they are about to proceed to M5. The figure depicts current process information transmitted to the HoloLens.

During each experiment, a referee recorded the TCT and NoE. After completing the assembly, participants filled out two questionnaires: SUS and NASA-TLX. SUS was used to measure usability while NASA-TLX was used to measure cognitive load. Additionally, participants were interviewed to rate the system from multiple perspectives, including usability, efficiency, feeling, attention, interaction effect, feasibility, and convenience. They also described the advantages and disadvantages of the system and selected the most useful type of information for themselves. An example of detailed data for each participant are shown in Fig. 11.

5.3.2. Results and discussion

To verify the validity of hypothesis H1 to H5, the results are analyzed and discussed in terms of 5 aspects including TCT, NoE, cognitive load, system usability, and user experience. Based on the critical requirements of the hypotheses, we selected method $\times$ skill and method $\times$ fatigue as two interaction groupings for two-way ANOVA analyses, allowing assessment of method performance across varying skill levels and fatigue states. To meet the assumptions for ANOVA, normality (Shapiro–Wilk test) and homogeneity of variance (Levene's test) checks were performed on each method $\times$ skill and method $\times$ fatigue combination, confirming normal distribution and homogeneity of variances. Two-way ANOVA analyses were subsequently conducted in IBM-SPSS to compare variances across groups and evaluate significant differences. Main effects of each variable were also examined independently to test the specific expectations of H1 to H5. Table 13 presents the mean TCT, NoE, NASA-TLX scores, and SUS scores of the 48 volunteers across different methods and skill levels, while Table 14 shows these mean scores across different methods and fatigue levels.

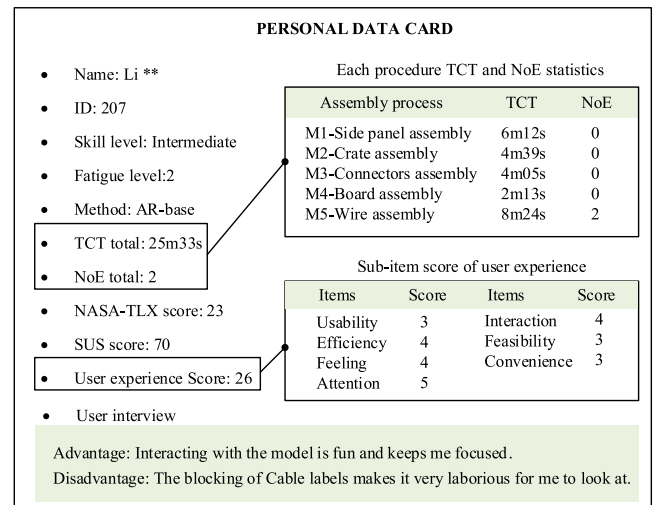


Fig. 11. Personal data representation.

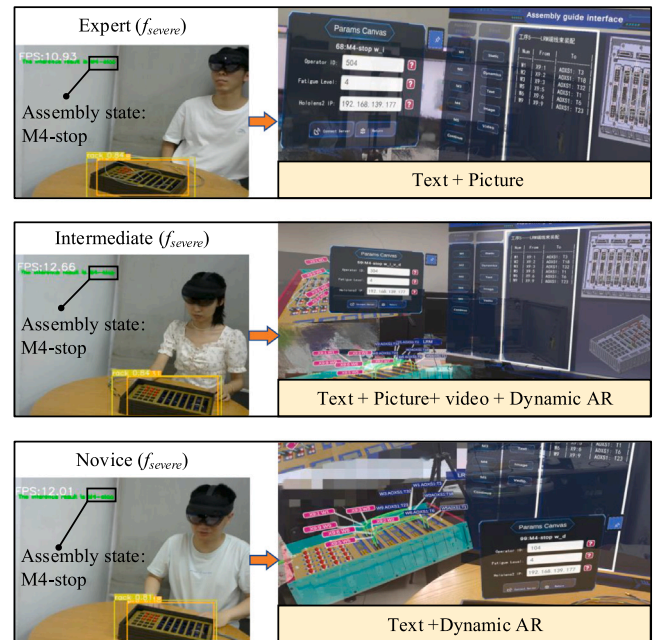


Fig. 12. Assembly process using AR-AIP for operators with varying skill levels under f_{severe} .

(a) Task Completion Time (TCT)

The two-way ANOVA for method and skill shows that both method and skill are significant factors influencing TCT. The main effect of method on TCT is significant ($F(1,42) = 13.518, p < 0.001$), with the average TCT for AR-AIP ($M = 24.079$) being notably lower than that for AR-Base ($M = 27.841$), representing a reduction of 13.51%. The main effect of skill level ($F(2,42) = 59.145, p < 0.001$) indicates that, across both methods, individuals with higher skill levels completed tasks significantly faster than those with lower skill levels. The interaction effect between skill and method ($F(2,42) = 3.267, p = 0.048$) is statistically significant. As shown in Fig. 13(a), experts using AR-AIP had an average time ($M = 19.468$) that was 6.29% shorter than AR-Base ($M = 20.774$), technicians using AR-AIP had an average time ($M = 23.014$) that was 10.13% shorter than AR-Base ($M = 25.609$), and novices using AR-AIP had an average time ($M = 29.755$) that

Table 13Assembly performance data of users under different methods and skills (Mean $\pm$ SD).

Method	Skill	TCT	NoE	NASA-TLX	SUS
AR-AIP	Expert	19.468 $\pm$ 4.014	0.750 $\pm$ 0.707	17.630 $\pm$ 2.326	82.500 $\pm$ 10.177
	Intermediate	23.014 $\pm$ 2.058	1.050 $\pm$ 1.069	22.380 $\pm$ 4.173	78.750 $\pm$ 8.864
	Novice	29.755 $\pm$ 2.854	2.880 $\pm$ 1.126	25.250 $\pm$ 3.882	85.313 $\pm$ 4.105
AR-Base	Expert	20.774 $\pm$ 3.634	1.000 $\pm$ 1.069	19.880 $\pm$ 2.167	65.000 $\pm$ 9.354
	Intermediate	25.609 $\pm$ 2.919	2.380 $\pm$ 0.744	24.880 $\pm$ 3.758	64.063 $\pm$ 9.722
	Novice	37.140 $\pm$ 5.016	4.630 $\pm$ 1.188	35.380 $\pm$ 6.046	68.750 $\pm$ 5.510
<i>F</i>		$F_m = 13.518^{**}$ $F_s = 59.145^{**}$ $F_{m \times s} = 3.267^{**}$	$F_m = 10.988^*$ $F_s = 33.712^{**}$	$F_m = 18.979^{**}$ $F_s = 34.684^{**}$ $F_{m \times s} = 5.456^*$	$F_m = 46.216^{**}$

* indicates $p < 0.05$, ** indicates $p < 0.001$. F_m indicates the F -value of main effect of methods, F_s indicates the F -value of main effect of skills, $F_{m \times s}$ indicates the F -value of the interaction effect of methods and skills.

Table 14Assembly performance data of users under different methods and fatigues (Mean $\pm$ SD).

Method	Fatigue	TCT	NoE	NASA-TLX	SUS
AR-AIP	1	21.598 $\pm$ 5.063	1.330 $\pm$ 0.816	17.670 $\pm$ 2.582	82.500 $\pm$ 12.748
	2	22.438 $\pm$ 5.272	1.500 $\pm$ 1.049	20.330 $\pm$ 3.386	80.000 $\pm$ 8.944
	3	25.760 $\pm$ 4.367	2.170 $\pm$ 1.472	22.670 $\pm$ 3.445	82.917 $\pm$ 5.792
	4	27.520 $\pm$ 4.404	1.830 $\pm$ 1.835	26.330 $\pm$ 4.676	83.333 $\pm$ 5.401
AR-Base	1	23.950 $\pm$ 6.867	2.170 $\pm$ 1.602	22.170 $\pm$ 5.115	63.333 $\pm$ 7.528
	2	25.468 $\pm$ 7.543	2.000 $\pm$ 1.414	24.830 $\pm$ 6.369	67.917 $\pm$ 8.429
	3	28.838 $\pm$ 6.663	2.830 $\pm$ 2.041	27.830 $\pm$ 7.414	67.500 $\pm$ 8.367
	4	33.107 $\pm$ 9.167	3.670 $\pm$ 2.066	32.000 $\pm$ 9.592	65.000 $\pm$ 10.124
<i>F</i>		$F_m = 4.188^*$ $F_f = 3.784^*$	$F_m = 4.343^*$	$F_m = 8.887^{**}$ $F_f = 5.631^{**}$	$F_m = 41.872^{**}$

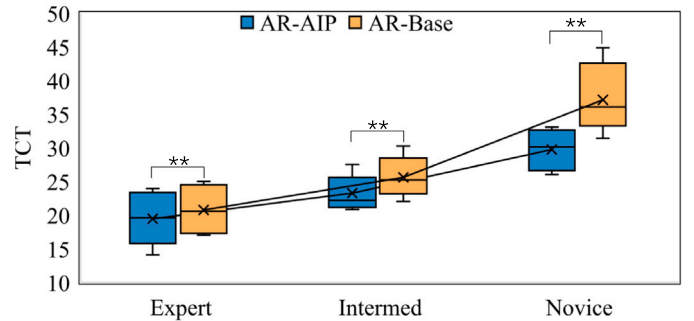
F_m indicates the F -value of main effect of methods, F_f indicates the F -value of main effect of fatigues.

was 19.88% shorter than AR-Base ($M = 37.140$). This indicates that the benefit of using AR-AIP is more pronounced for novices, and the AR-AIP method can compensate for skill differences, leading to more consistent efficiency across different skill levels. This result provides partial support for Hypothesis 1.

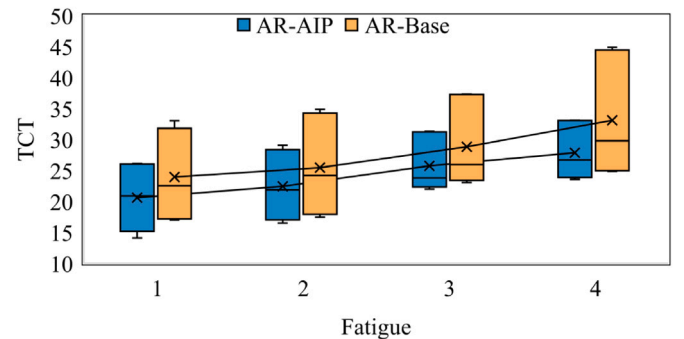
The two-way ANOVA for method and fatigue indicates that both method and fatigue are significant factors influencing TCT. Specifically, the method has a significant effect on TCT ($F(1,40) = 4.188, p = 0.047$), which aligns with the results of the two-way ANOVA for method $\times$ skill. The main effect of fatigue ($F(3,40) = 3.784, p = 0.018$) is also significant, showing that different levels of fatigue significantly impact TCT. Although the interaction effect between method and fatigue was not significant, as shown in Fig. 13(b), the TCT under AR-AIP was reduced by 9.8%, 11.90%, 10.67%, and 16.88% at fatigue levels 1, 2, 3, and 4, respectively, compared to AR-Base. This indicates that using the AR-AIP method can mitigate the impact of fatigue on TCT. This finding suggests that personalizing information combinations based on users' fatigue levels can reduce the negative effects of fatigue on operational efficiency. Notably, in states of severe fatigue, the AR-AIP method showed a more pronounced effect, effectively reducing the fatigue-related decline in work efficiency. Overall, AR-AIP reduces the variability in TCT caused by differences in skill levels and fatigue, highlighting its advantages in managing task efficiency under variable conditions. Therefore, the results support Hypothesis 1.

(b) Number of Errors (NoE)

The two-way ANOVA results for method and skill indicate that both method ($F(1,42) = 10.988, p < 0.05$) and skill level ($F(2,42) = 33.712, p < 0.001$) are significant factors influencing the NoE. Specifically, the AR-AIP group consistently demonstrated fewer errors than the AR-Base group, as skill level decreased, the number of errors progressively increased. However, the interaction effect between skill level and method was not statistically significant ($p > 0.05$). Further analysis, as shown in Fig. 14, reveals that, compared to experts and intermediates, novices in the AR-AIP group had significantly lower



(a) Interaction effect of Method and Skill

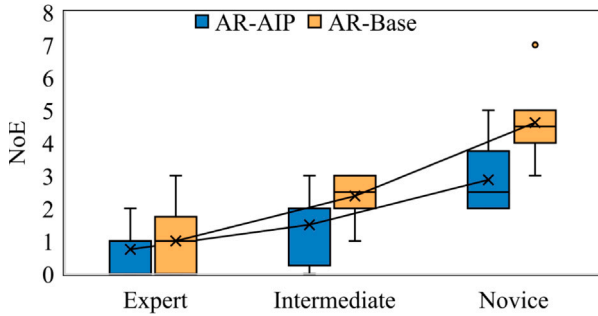


(b) Interaction effect of Method and Fatigue

Fig. 13. Average task completion time for different factors.

Table 15User experiences under different methods, skills, and fatigues (Mean $\pm$ SD).

Influence factors		Usability	Efficiency	Feeling	Attention	Interaction	Feasibility	Convenience	Total
Method	AR-AIP	3.83 $\pm$ 0.702	4.33 $\pm$ 0.565	3.92 $\pm$ 0.654	4.38 $\pm$ 0.774	3.33 $\pm$ 0.816	4.58 $\pm$ 0.654	4.12 $\pm$ 0.682	28.17 $\pm$ 2.426
	AR-Base	3.58 $\pm$ 0.776	3.46 $\pm$ 0.588	3.29 $\pm$ 0.751	4.17 $\pm$ 0.816	3.71 $\pm$ 0.751	4.21 $\pm$ 0.721	3.63 $\pm$ 0.771	26.29 $\pm$ 2.629
Skill	Expert	3.69 $\pm$ 0.704	3.94 $\pm$ 0.854	3.50 $\pm$ 0.816	4.38 $\pm$ 0.719	3.62 $\pm$ 0.885	4.56 $\pm$ 0.682	3.94 $\pm$ 0.682	27.25 $\pm$ 2.978
	Intermediate	3.88 $\pm$ 0.719	4.13 $\pm$ 0.619	3.69 $\pm$ 0.802	4.25 $\pm$ 0.856	3.50 $\pm$ 0.730	4.56 $\pm$ 0.612	3.94 $\pm$ 0.682	27.25 $\pm$ 2.975
	Novice	3.56 $\pm$ 0.814	3.62 $\pm$ 0.619	3.63 $\pm$ 0.885	4.19 $\pm$ 0.834	3.44 $\pm$ 0.814	4.06 $\pm$ 0.854	3.75 $\pm$ 0.775	26.52 $\pm$ 2.530
Fatigue	f_{non}	3.58 $\pm$ 0.515	3.67 $\pm$ 0.651	3.75 $\pm$ 0.452	4.50 $\pm$ 0.522	3.58 $\pm$ 0.996	4.58 $\pm$ 0.669	4.08 $\pm$ 0.900	27.92 $\pm$ 1.834
	f_{mild}	3.83 $\pm$ 0.718	4.08 $\pm$ 0.793	3.75 $\pm$ 0.754	4.67 $\pm$ 0.651	3.58 $\pm$ 0.669	4.42 $\pm$ 0.515	3.92 $\pm$ 0.515	27.92 $\pm$ 2.575
	$f_{moderate}$	3.58 $\pm$ 0.669	3.83 $\pm$ 0.835	3.67 $\pm$ 0.778	4.67 $\pm$ 0.651	3.25 $\pm$ 0.622	4.25 $\pm$ 0.754	3.67 $\pm$ 0.515	26.67 $\pm$ 2.498
	f_{severe}	3.83 $\pm$ 1.030	4.00 $\pm$ 0.603	3.67 $\pm$ 0.888	4.08 $\pm$ 0.900	3.71 $\pm$ 0.751	4.08 $\pm$ 0.900	3.83 $\pm$ 0.718	27.33 $\pm$ 3.750
<i>F</i>		/	$F_m = 25.940^{**}$	$F_m = 9.783^*$	$F_f = 3.347^*$	/	/	$F_m = 8.000^*$	$F_m = 6.756^*$

**Fig. 14.** Interaction effect of methods and skills.

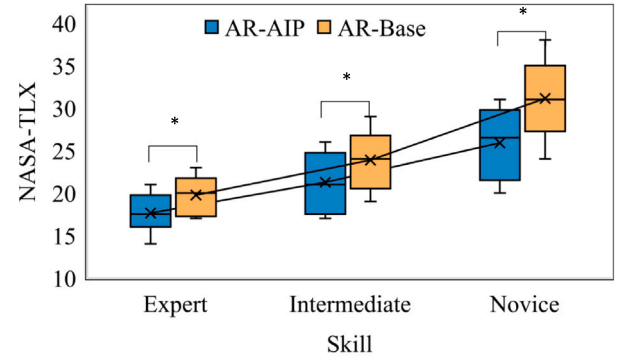
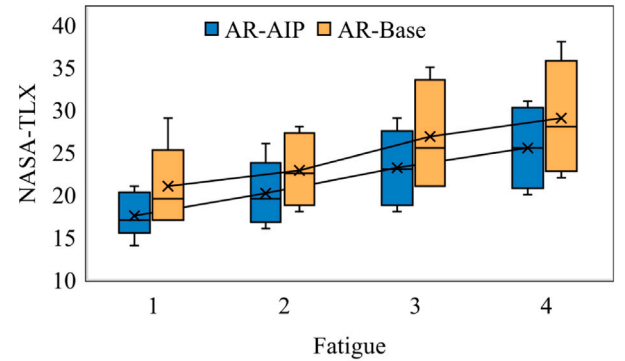
error rates than those in the AR-Base group. This finding strongly suggests that AR-AIP is more effective in reducing error rates among novices. By narrowing the error rate gap between novices and experienced operators, AR-AIP significantly enhances the accuracy of novice performance.

In addition, two-way ANOVA results for method and fatigue indicate that only the method ($F(1, 40) = 4.343$, $p < 0.05$) has a significant impact on the NoE. The main effect of fatigue was not significant ($p > 0.05$), nor was the interaction effect between method and fatigue. These findings suggest that AR-AIP can help operators reduce errors while also mitigating some of the error variability associated with different skill levels. Therefore, Hypothesis 2 is partially supported.

(c) Cognitive Load

Cognitive load is a crucial indicator for evaluating the effectiveness of a method, assessing the total amount of cognitive resources required to complete a task. NASA-TLX is used to evaluate the impact of different methods on the cognitive load of operators. Higher NASA-TLX score indicates heavier cognitive load. The two-way ANOVA results for method and skill indicate that both method ($F(1, 42) = 18.979$, $p < 0.05$) and skill level ($F(2, 42) = 34.684$, $p < 0.001$) are significant factors influencing the cognitive load. The AR-AIP group consistently exhibited lower cognitive load after task completion compared to the AR-Base group. As depicted in Fig. 15(a), there is an interaction effect between method and skill ($F(2, 42) = 5.456$, $p < 0.05$). Under the influence of AR-AIP, cognitive load levels across different skill levels were significantly lower than those in the AR-Base group. This effect was particularly pronounced among novices, where the cognitive load was significantly reduced compared to both experts and intermediates. The similarity in cognitive load levels among these three groups suggests that AR-AIP mitigates the variability in cognitive load caused by differences in skill levels.

The two-way ANOVA for method and fatigue indicates that both method and fatigue are significant factors influencing the cognitive load. Specifically, the method has a significant effect on the cognitive load ($F(1, 40) = 8.887$, $p < 0.05$), which aligns with the results of the two-way ANOVA for method $\times$ skill. The main effect of fatigue

**(a)** Interaction effect of Method and Skill**(b)** Interaction effect of Method and Fatigue**Fig. 15.** Average NASA-TLX score for different factors.

($F(3, 40) = 5.631$, $p < 0.05$) is also significant, showing that different levels of fatigue significantly impact the cognitive load. The interaction between fatigue and method was not statistically significant ($p > 0.05$). Combining this with Fig. 15(b), it becomes evident that the influence of AR-AIP on cognitive load remains almost consistent across all fatigue levels. There appears to be no significant effort to narrow the gap between these levels caused by fatigue. Therefore, hypothesis 3 is partially accepted.

(d) System Usability

Both two-way ANOVAs for method and skill, and for method and fatigue, indicate that only the method was a significant factor influencing system usability, as shown by the results in Table 13 ($F(1, 42) = 46.216$, $p < 0.001$) and Table 14 ($F(1, 40) = 41.872$, $p < 0.001$). As shown in Fig. 16, users rated AR-AIP ($M = 82.188$) significantly higher than AR-Base ($M = 65.938$). The experimental result indicates that the AR-AIP group provides better system usability, thereby accepting hypothesis 4.

(e) User Experience

After completing the assembly tasks, participants were interviewed and rated the system in terms of different factors including usability,

Table 16
Feedbacks of users interviewed.

	AR-Base	AR-AIP
Disadvantages	• The interface and model operations are challenging, and button clicking is cumbersome. • Information can be redundant.	• High demands on movements, otherwise incorrect information is pushed. • Misidentifications sometimes cause other windows to pop up, requiring manual correction by clicking buttons.
Advantages	• Cable model labels obstruct the view. • Buttons are easily accidentally touched during assembly operations. • Prolonged use causes discomfort around the eyes, severe heating, and system lag. • The field of view is too small, limiting the effectiveness of interactions.	• No need for button operations, making it highly suitable for manual tasks. • The information pushed is timely, and the combination of various types of information clarifies tasks quickly.
	• Interacting with the interface and models is enjoyable. • The AR models are well-made, with labels that help quickly locate parts.	
	• Suitable for novices to learn. • Dynamic AR models (animations) are very helpful for assembly operations.	

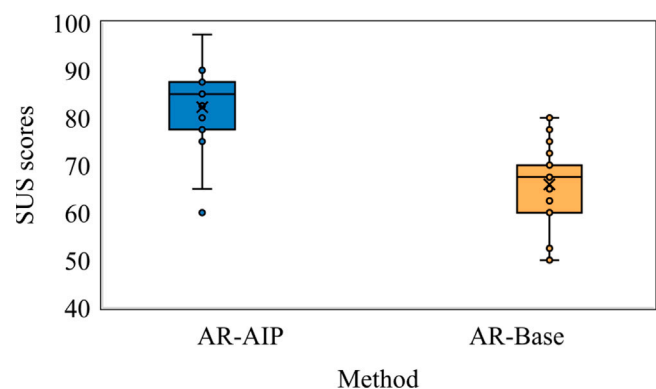


Fig. 16. Average SUS scores for different methods.

efficiency, feeling, attention, interaction effect, feasibility, and convenience. As shown in Table 15, the overall analysis indicated that only the method had a statistically significant main effect on the total score ($F(1, 24) = 6.756, p = 0.016$). The mean score for AR-AIP was significantly higher than for AR-Base, particularly in usability, efficiency, feeling, feasibility, and convenience, indicating a higher overall satisfaction with the AR-AIP system (Fig. 17). The user experience ratings of AR-AIP group under fatigue conditions are slightly lower than non-fatigue conditions, but the difference is not significant. However, in terms of interaction, AR-AIP scores slightly lower than AR-Base, despite both methods receiving criticisms in this area (see Table 16). The AR-Base group involves more human-interface interaction such as clicking button, which may stimulate their interest. In contrast, AR-AIP minimizes this interaction to free hands and improve work efficiency. Overall, AR-AIP contributes to improved operator satisfaction and efficiency, thereby hypothesis 5 is supported.

In addition to the ratings, participants were also interviewed about the advantages and disadvantages of the two methods, as recorded in Table 16. The AR-Base method received criticism for its challenging interface and model manipulation, particularly the cumbersome nature of button-clicking operations. Users found the abundance of information overwhelming, especially for highly skilled individuals who are difficult to concentrate on critical details. In contrast, the AR-AIP method avoided explicit interaction problems. However, it has strict requirements on the user's actions, and occasionally there will be the problem of incorrect information push caused by inaccurate actions, which needs to be corrected by manually clicking the button.

Common drawbacks found for both methods included the view obstruction, accidental button clicking, and hardware-related issues

such as overheating and lag, which negatively impacted the field of view and interaction effectiveness. These hardware constraints were noted as the most recurrent drawback, causing eye discomfort and limiting the user experience.

On the positive side, the AR-Base users appreciated the interactive interface and models, expressing enjoyment and interest in the AR environment. The labeling of parts was particularly useful for quick identification, and the capability to manipulate holographic models from various angles was seen as advantageous. Meanwhile, the AR-AIP group highlighted the absence of redundant button clicking and the efficient push of timely information, which helped users to quickly comprehend tasks and maintain focus. Dynamic AR models and animations were also commended for their assistance in assembly operations.

6. Conclusions

This study proposes an intelligent AR-guided adaptive information push (AR-AIP) method based on context-awareness. The method can perceive human cognitive ability and assembly state in real time, and adjust the transmission of assembly guidance information based on information load. Using the case of electronic equipment rack assembly, this research first experimentally verifies the classification of fatigue states, then validates the assembly state perception method, and finally conducts comprehensive system experiments with participants. The results show that AR-AIP offers greater user-friendliness compared to traditional AR-guided assembly. It helps improve efficiency, reduce errors, and ease cognitive load. Additionally, AR-AIP demonstrates an advantage in minimizing user differences caused by skill levels and fatigue impacts. It can adaptively provide appropriate information to individuals of different skill levels and fatigue states, significantly enhancing work efficiency and accuracy, particularly benefiting novices and individuals under high fatigue intensity. This approach provides a new technological pathway for intelligent manufacturing and human-machine collaboration, with promising industrial applications and considerable value.

However, this study still has some limitations. Currently, the occlusion detection and precise sensing of assembly states for certain small components, such as SMA connectors, require further optimization. Additionally, due to sample size constraints, only two-factor interaction analyses were performed for selected factors. Future research could expand the sample size to explore complex interactions among skill levels, fatigue, and other factors. Longitudinal studies could also be considered to examine the effects of AR-AIP during prolonged use in industrial settings, while further exploring the individual roles of context awareness and information load matching in AR-AIP. Through targeted experimental designs, we will analyze the contributions of

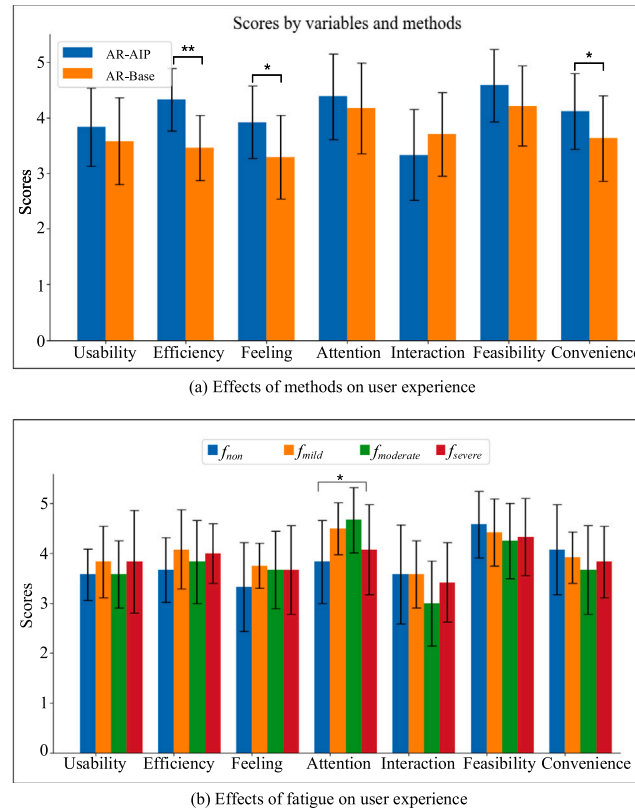


Fig. 17. Average user experience scores for different factors.

each component to the overall system performance, providing more focused data support for the enhancement of AR-AIP.

While this study represents an initial exploration into AR information load quantification and push, insights gained from participant interviews suggest broader applications beyond assembly guidance, including novice training and maintenance guidance. Future research directions include expanding AR visualization forms tailored to specific assembly scenarios, utilizing generative AI to explore more diverse and effective AR content authoring methods, integrating diverse human-computer interaction modes, and adaptively delivering AR process information carriers based on operator cognitive capabilities. These efforts aim to empower manual assembly processes with more flexible AR technology solutions.

CRediT authorship contribution statement

Binbin Wang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Lianyu Zheng:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization. **Yiwei Wang:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Lihui Wang:** Writing – review & editing, Supervision, Conceptualization. **Zhonghua Qi:** Writing – review & editing, Validation, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. List of symbols

List of symbols

$Act = \{act_1, act_2, \dots, act_i, \dots, act_m\}$	Set of assembly actions in an assembly process.
act_i	Subsets of the sets Act , representing all the actions involved in a procedure.
$C = \{c_1, c_2, \dots, c_i, \dots, c_J\}$	Set of all core process elements.
$C_A = \{skill, fatigue\}$	Cognitive ability
c_i	i th type of core elements.
c_{ij}, c_{kl}	Core element entity.
D_c	Dimension of core elements, either 2D or 3D.
D_e	Dimension of representation forms, either 2D or 3D.
$E = \{e_1, e_2, \dots, e_j, \dots, e_J\}$	Set of all AR representation forms.
$Ea = [ea_{ki}]_{K \times I}$	Expression ability matrix ($K \times I$).

$ea_{ki} = \sum_{j \in E_k}^{N_k} ef_{ij} \times ee_{ij}$	The element in the matrix Ea . ea_{ki} measures the extent to which representation group RG_k can effectively and clearly express the core element c_i .	$M = \{M_1, M_2, \dots, M_k, \dots, M_l\}$	Set of assembly procedures in assembly process.
Ed = $[ed_{ki}]_{K \times I}$ $ed_{ki} = [\text{round}(ea_{ki} - tc_i)] + 1$	Expression intensity matrix ($K \times I$). The element in the matrix Ed . ed_{ki} quantifies how clearly representation group RG_k can express the core element c_i under specific task complexity.	m_i	The number of entity classes that the i th core element type has in procedure M_k .
Ee = $[ee_{ij}]_{I \times J}$ $ee_{ij} = \frac{1}{Ia_{ij}}$	Expression efficacy matrix ($I \times J$). The element in the matrix Ee . ee_{ij} indicates the degree to which representation form e_j can clearly express the core element c_i .	M_k	k th assembly procedure in an assembly process.
Ef = $[ef_{ij}]_{I \times J}$ ef_{ij}	Expression feasibility matrix ($I \times J$). The element in the matrix Ef . ef_{ij} refers to whether the representation form e_j can clearly express the core element c_i .	mRMR	The Maximum Relevance Minimum Redundancy.
e_j $E_k = \{j e_{k,n} = e_j\}$	j th type of AR representation forms. Set of indices j of AR representation forms included in the representation group RG_k .	$N = \sum_{i=1}^I \sum_{j=1}^{m_i} n_{ij}$	The total number of occurrences of all core element entities.
$e_{k,n}$	n th representation form included in the k th representation group.	n_{ij}	The number of occurrences of core element entity c_{ij} .
$fatigue$	The user's fatigue level, including: f_{non} , f_{mild} , $f_{moderate}$ and f_{severe} four levels.	N_k	Number of AR representation forms included in the k th representation group RG_k .
$f_{ij} = n_{ij} / N$	The frequency of occurrence of core element entity c_{ij} .	$Part = \{part_1, part_2, \dots, part_j, \dots, part_n\}$ $part_j$	Set of assembly parts in an assembly process. Subsets of the sets $Part$, representing all the parts involved in a procedure.
$f_{non}, f_{mild}, f_{moderate}, f_{severe}$	Non-fatigue, Mild-fatigue, Moderate-fatigue, Severe-fatigue.	$p_{ij} = L_{ij} / L$	The probability of understanding the relationship between a core element entity c_{ij} and other core element entities.
$H_{AR} = H_p + H_u + H_t$	AR assembly process information load.	$p_q = 1 / ed_{k,i}$	The probability that an operator can correctly understand the conveyed process information.
$H_k = g(M_k, C_A, Ea_k, H_{AR})$	AR guided assembly process information of the current procedure in adaptive push. The content presented is one of the representation group.	$p(x_i), p(x_j)$ $p(x_i, x_j)$ $q_{ij} = (Se + \alpha De) / (Sc + \alpha Dc)$	The edge probability density. The joint probability distribution. The probability that representation form e_j can express core element c_i .
$H_p = - \sum_{i=1}^I \sum_{j=1}^{m_i} f_{ij} \cdot \log_2 f_{ij}$	Information perception load.	$R(c_{ij}, c_{kl})$	The relationship between the core element entity c_{ij} and the core element entity c_{kl} .
$H_t = - \sum_{i=1}^I \sum_{q=1}^{ed_{ki}} p_q \cdot \log_2 p_q$	Information transmission load.	$RG_k = \{e_{k,n} n = 1, 2, \dots, N_k\}$	Set of AR representation forms included in the k th representation group.
$H_u = - \sum_{i=1}^I \sum_{j=1}^{m_i} p_{ij} \cdot \log_2 p_{ij}$	Information understanding load.	S_c	State (static or dynamic) of core elements.
$Ia_{ij} = -\log_2 q_{ij}$	The information content.	S_e	State of representation forms, either static or dynamic.
$I(X_i; X_j) = \sum_{x_i \in X_i} \sum_{x_j \in X_j} p(x_i, x_j) \log_2 \left(\frac{p(x_i, x_j)}{p(x_i)p(x_j)} \right)$	The redundancy between the features.	$S_{RG} = \{RG_1, RG_2, \dots, RG_k, \dots, RG_K\}$	Set of representation groups. The representation group is based on text and combined with the other four representation forms, resulting in a total of 15 combinations, i.e., text + image, text + video, text + static AR model, text + dynamic AR model, and so on.
$I(X_i; Y)$	The correlation between the feature and the label.	$Tc = \sum_{i=1}^I tc_i$ $tc_i = \sum_{d=1}^D (r_d \cdot w_d)$	Task complexity of a procedure. Complexity of a single core element, r_d and w_d represent complexity levels and weights, respectively.
$L = \sum_{i=1}^I \sum_{j=1}^{m_i} L_{ij}$	The total sum of relationships among all core element entities.	$uea_k = Ea_k / H_{AR}$	Expression ability per unit information load of the k th representation group.
$L_{ij} = \sum_{k=1}^I \sum_{l=1}^{m_i} R(c_{ij}, c_{kl})$	The sum of relationships between the core element entity c_{ij} and all other core element entities (including itself).	X_i, X_j Y	Brain wave signal characteristic. Label for classification of brainwave signals.
L_{t-1}, L_t	Video list of action recognition.	$\alpha = 0.8$	A parameter used to distinguish the importance of dimension and state.

Appendix B. MFI-20 questionnaire

Multidimensional Fatigue Inventory (MFI) questionnaire

Name: _____ ID: _____ Gender: _____ Age: _____ Date: _____

Instructions:

By means of the following statements we would like to get an idea of how you have been feeling **now**. There is, for example, the statement:

"I FEEL RELAXED"

If you think that this is **entirely true**, that indeed you have been feeling relaxed **now**, please, place a **✓** in the extreme left box; like this:

no, that is not true ☐1 ☐2 ☐3 ☐4 ☒5 yes, that is true

The more you **disagree** with the statement, the more you can place a **✓** in the direction of "no, that is not true". Please do not miss out a statement and place only one **✓** in a box for each statement.

1	I feel fit.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
2	Physically, I feel only able to do a little.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
3	I feel very active.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
4	I feel like doing all sorts of nice things.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
5	I feel tired.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
6	I think I do a lot in a day.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
7	When I am doing something, I can keep my thoughts on it.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
8	Physically I can take on a lot.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
9	I dread having to do things.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
10	I think I do very little in a day.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
11	I can concentrate well.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
12	I am rested.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
13	It takes a lot of effort to concentrate on things.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
14	Physically I feel I am in a bad condition.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
15	I have a lot of plans.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
16	I tire easily.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
17	I get little done.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
18	I don't feel like doing anything.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
19	My thoughts easily wander.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true
20	Physically I feel I am in an excellent condition.	no, that is not true	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	yes, that is true

Thank you very much for your cooperation!

Data availability

Data will be made available on request.

References

[1] Y. Yang, J. Karreman, M. de Jong, Comparing the effects of paper and mobile augmented reality instructions to guide assembly tasks, in: 2020 IEEE International Professional Communication Conference, ProComm, 2020, pp. 96–104, <http://dx.doi.org/10.1109/ProComm48883.2020.00021>.

[2] B. Wang, L. Zheng, Y. Wang, W. Fang, L. Wang, Towards the industry 5.0 frontier: Review and prospect of XR in product assembly, J. Manuf. Syst. 74 (2024) 777–811, <http://dx.doi.org/10.1016/j.jmsy.2024.05.002>.

[3] R.G. Boboc, F. Gîrbacia, E.V. Butilă, The application of augmented reality in the automotive industry: A systematic literature review, Appl. Sci. 10 (12) (2020) <http://dx.doi.org/10.3390/app10124259>.

[4] M. Funk, T. Kosch, A. Schmidt, Interactive worker assistance: comparing the effects of in-situ projection, head-mounted displays, tablet, and paper instructions, in: Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing, 2016, pp. 934–939, <http://dx.doi.org/10.1145/2971648.2971706>.

[5] L.F. de Souza Cardoso, F.C.M.Q. Mariano, E.R. Zorzal, A survey of industrial augmented reality, Comput. Ind. Eng. 139 (2020) 106159, <http://dx.doi.org/10.1016/j.cie.2019.106159>.

[6] J. Liu, M.J. Land, J.A.C. Bokhorst, Q. Chen, Improving coordination in assembly job shops: redesigning order release and dispatching, Flex. Serv. Manuf. J. 35 (3) (2023) 669–697, <http://dx.doi.org/10.1007/s10696-022-09454-x>.

- [7] X. Wang, S. Ong, A. Nee, Multi-modal augmented-reality assembly guidance based on bare-hand interface, *Adv. Eng. Inform.* 30 (3) (2016) 406–421, <http://dx.doi.org/10.1016/j.aei.2016.05.004>.
- [8] D. Bläsing, S. Hinrichsen, M. Bornewasser, Reduction of cognitive load in complex assembly systems, in: *Human Interaction, Emerging Technologies and Future Applications II*, Springer, 2020, pp. 495–500.
- [9] A. Peters, B.S. McEwen, K. Friston, Uncertainty and stress: Why it causes diseases and how it is mastered by the brain, *Prog. Neurobiol.* 156 (2017) 164–188, <http://dx.doi.org/10.1016/j.pneurobio.2017.05.004>.
- [10] J.M. Driscoll, M.J. Corpolongo, Uncertainty estimation and the uncertainty-probability shift under information load, *Syst. Res. Behav. Sci.* 25 (1980) 205–218, <http://dx.doi.org/10.1002/bs.3830250305>.
- [11] P.M. Allen, J.A. Edwards, F.J. Snyder, K.A. Makinson, D.M. Hamby, The effect of cognitive load on decision making with graphically displayed uncertainty information, *Risk Anal.* 34 (8) (2014) 1495–1505, <http://dx.doi.org/10.1111/risa.12161>.
- [12] R. Radkowski, J. Herrema, J.H. Oliver, Augmented reality-based manual assembly support with visual features for different degrees of difficulty, *Int. J. Hum.-Comput. Interact.* 31 (2015) 337–349, <http://dx.doi.org/10.1080/10447318.2014.994194>.
- [13] G. Zhao, P. Feng, J. Zhang, D. Yu, Z. Wu, Information integration and instruction authoring of augmented assembly systems, *Int. J. Intell. Syst.* 36 (9) (2021) 5028–5050, <http://dx.doi.org/10.1002/int.22501>.
- [14] J. Zhang, S. Wang, W. He, J. Li, Z. Cao, B. Wei, Projected augmented reality assembly assistance system supporting multi-modal interaction, *Int. J. Adv. Manuf. Technol.* 123 (3) (2022) 1353–1367, <http://dx.doi.org/10.1007/s00170-022-10113-6>.
- [15] Z. Lai, W. Tao, M.C. Leu, Z. Yin, Smart augmented reality instructional system for mechanical assembly towards worker-centered intelligent manufacturing, *J. Manuf. Syst.* 55 (2020) 69–81, <http://dx.doi.org/10.1016/j.jmsy.2020.02.010>.
- [16] E. Lampen, J. Lehwald, T. Pfeiffer, A context-aware assistance framework for implicit interaction with an augmented human, in: *Virtual, Augmented and Mixed Reality. Industrial and Everyday Life Applications*, Springer, 2020, pp. 91–110, http://dx.doi.org/10.1007/978-3-030-49698-2_7.
- [17] J. Lee, T. Lim, W. Kim, Investigating the usability of collaborative robot control through hands-free operation using eye gaze and augmented reality, in: *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS, 2023*, pp. 4101–4106, <http://dx.doi.org/10.1109/IROS55552.2023.10342045>.
- [18] S.O. J. Zhu, A. Nee, A context-aware augmented reality assisted maintenance system, *Int. J. Comput. Integr. Manuf.* 28 (2) (2015) 213–225, <http://dx.doi.org/10.1080/0951192X.2013.874589>.
- [19] D. Moutinho, L. F. Rocha, C.M. Costa, L.F. Teixeira, G. Veiga, Deep learning-based human action recognition to leverage context awareness in collaborative assembly, *Robot. Comput.-Integr. Manuf.* 80 (2023) 102449, <http://dx.doi.org/10.1016/j.rcim.2022.102449>.
- [20] A. Dasgupta, M. Manuel, R.S. Mansur, N. Nowak, D. Gračanin, Towards real time object recognition for context awareness in mixed reality: A machine learning approach, in: *2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops, VRW, 2020*, pp. 262–268, <http://dx.doi.org/10.1109/VRW50115.2020.00054>.
- [21] Z. Yang, J. Shi, W. Jiang, Y. Sui, Y. Wu, S. Ma, C. Kang, H. Li, Influences of augmented reality assistance on performance and cognitive loads in different stages of assembly task, *Front. Psychol.* 10 (2019) <http://dx.doi.org/10.3389/fpsyg.2019.01703>.
- [22] P. Vanneste, Y. Huang, J.Y. Park, F. Cornillie, B. Declodet, W.V. den Noortgate, Cognitive support for assembly operations by means of augmented reality: an exploratory study, *Int. J. Hum.-Comput. Stud.* 143 (2020) 102480, <http://dx.doi.org/10.1016/j.ijhcs.2020.102480>.
- [23] J. Van Cutsem, S. Marcora, K. De Pauw, S. Bailey, R. Meeusen, B. Roelands, The effects of mental fatigue on physical performance: A systematic review, *Sports Med.* 47 (8) (2017) 1569–1588, <http://dx.doi.org/10.1007/s40279-016-0672-0>.
- [24] J.E. Hunter, Cognitive ability, cognitive aptitudes, job knowledge, and job performance, *J. Vocat. Behav.* 29 (3) (1986) 340–362, [http://dx.doi.org/10.1016/0001-8791\(86\)90013-8](http://dx.doi.org/10.1016/0001-8791(86)90013-8).
- [25] K. Nishida, M. Itoh, K. Nakamoto, Augmented reality-based system for skill transfer of workpiece fixturing in turning operations, *Int. J. Autom. Technol.* 17 (2) (2023) 136–143, <http://dx.doi.org/10.20965/ijat.2023.p0136>.
- [26] Y. Yan, X. Bai, W. He, S. Wang, X. Zhang, P. Wang, L. Liu, Q. Yu, A novel adaptive visualization method based on user intention in ar manual assembly, *Int. J. Adv. Manuf. Technol.* 129 (9) (2023) 4705–4730, <http://dx.doi.org/10.1007/s00170-023-12557-w>.
- [27] W.D. Killgore, Effects of sleep deprivation on cognition, in: *Progress in Brain Research*, Vol. 185, Elsevier, 2010, pp. 105–129, <http://dx.doi.org/10.1016/B978-0-444-53702-7.00007-5>.
- [28] A. Othmani, A.Q.M. Sabri, S. Aslan, F. Chaieb, H. Rameh, R. Alfred, D. Cohen, EEG-based neural networks approaches for fatigue and drowsiness detection: A survey, *Neurocomputing* 557 (2023) 126709, <http://dx.doi.org/10.1016/j.neucom.2023.126709>.
- [29] C. Messeri, A. Bicchì, A.M. Zanchettin, P. Rocco, A dynamic task allocation strategy to mitigate the human physical fatigue in collaborative robotics, *IEEE Robot. Autom. Lett.* 7 (2) (2022) 2178–2185, <http://dx.doi.org/10.1109/LRA.2022.3143520>.
- [30] S.K. Ehrlich, G. Cheng, Human-agent co-adaptation using error-related potentials, *J. Neural Eng.* 15 (6) (2018) 066014, <http://dx.doi.org/10.1088/1741-2552/aae069>.
- [31] X. Liu, L. Zheng, Y. Wang, W. Yang, Z. Jiang, B. Wang, F. Tao, Y. Li, Human-centric collaborative assembly system for large-scale space deployable mechanism driven by digital twins and wearable AR devices, *J. Manuf. Syst.* 65 (2022) 720–742, <http://dx.doi.org/10.1016/j.jmsy.2022.11.007>.
- [32] X.L. Min Cai, C. Liu, Task allocation strategies considering task matching and ergonomics in the human-robot collaborative hybrid assembly cell, *Int. J. Prod. Res.* 61 (21) (2023) 7213–7232, <http://dx.doi.org/10.1080/00207543.2022.2147234>.
- [33] L.F. Austria-Melo, J. Cuellar-Castillo, A.A. Hernández, C. Montiel, D.A. Fabila-Bustos, M. Hernández-Chávez, Comparison of development and characteristics of several educational tools in augmented reality for visualization of 3D models difficult to understand. Chemistry application case, in: *2022 IEEE Mexican International Conference on Computer Science, ENC, 2022*, pp. 1–7, <http://dx.doi.org/10.1109/ENC56672.2022.9882948>.
- [34] W. Li, J. Wang, S. Lan, Content authoring of augmented reality assembly process, *Comput. Integr. Manuf. Syst.* 25 (7) (2019) 1676–1684, <http://dx.doi.org/10.13196/j.cims.2019.07.008>.
- [35] W. Li, J. Wang, S. Jiao, M. Wang, S. Li, Research on the visual elements of augmented reality assembly processes, *Virtual Real. Intell. Hardw.* 1 (6) (2019) 622–634, <http://dx.doi.org/10.1016/j.vrih.2019.09.006>.
- [36] Z. Wang, X. Bai, S. Zhang, M. Billingham, W. He, Y. Wang, D. Han, G. Chen, J. Li, The role of user-centered AR instruction in improving novice spatial cognition in a high-precision procedural task, *Adv. Eng. Inform.* 47 (2021) 101250, <http://dx.doi.org/10.1016/j.aei.2021.101250>.
- [37] Z. Wang, X. Bai, S. Zhang, W. He, X. Zhang, L. Zhang, P. Wang, D. Han, Y. Yan, Information-level AR instruction: a novel assembly guidance information representation assisting user cognition, *Int. J. Adv. Manuf. Technol.* 106 (1) (2020) 603–626, <http://dx.doi.org/10.1007/s00170-019-04538-9>.
- [38] G. Phillips-Wren, M. Adya, Decision making under stress: the role of information overload, time pressure, complexity, and uncertainty, *J. Decis. Syst.* 29 (sup1) (2020) 213–225, <http://dx.doi.org/10.1080/12460125.2020.1768680>.
- [39] Z. Wang, X. Bai, S. Zhang, M. Billingham, W. He, P. Wang, W. Lan, H. Min, Y. Chen, A comprehensive review of augmented reality-based instruction in manual assembly, training and repair, *Robot. Comput.-Integr. Manuf.* 78 (2022) 102407, <http://dx.doi.org/10.1016/j.rcim.2022.102407>.
- [40] L. Zheng, X. Liu, Z. An, S. Li, R. Zhang, A smart assistance system for cable assembly by combining wearable augmented reality with portable visual inspection, *Virtual Real. Intell. Hardw.* 2 (1) (2020) 12–27, <http://dx.doi.org/10.1016/j.vrih.2019.12.002>.
- [41] S. Li, P. Zheng, L. Zheng, An AR-assisted deep learning-based approach for automatic inspection of aviation connectors, *IEEE Trans. Ind. Inform.* 17 (3) (2021) 1721–1731, <http://dx.doi.org/10.1109/TII.2020.3000870>.
- [42] M. Fu, W. Fang, S. Gao, J. Hong, Y. Chen, Edge computing-driven scene-aware intelligent augmented reality assembly, *Int. J. Adv. Manuf. Technol.* 119 (11) (2022) 7369–7381, <http://dx.doi.org/10.1007/s00170-022-08758-4>.
- [43] W. Fang, T. Zhang, Z. Wang, J. Ding, A multi-modal context-aware sequence stage validation for human-centric AR assembly, *Comput. Ind. Eng.* 194 (2024) 110355, <http://dx.doi.org/10.1016/j.cie.2024.110355>.
- [44] C. Lv, B. Liu, D. Wu, J. Lv, J. Li, J. Bao, AR-assisted assembly method based on instance segmentation, *Int. J. Comput. Integr. Manuf.* (2024) 1–17, <http://dx.doi.org/10.1080/0951192X.2024.2328045>.
- [45] J. Dong, Z. Xia, Q. Zhao, Augmented reality assisted assembly training oriented dynamic gesture recognition and prediction, *Appl. Sci.* 11 (21) (2021) <http://dx.doi.org/10.3390/app11219789>.
- [46] X. Yin, X. Fan, W. Zhu, R. Liu, Synchronous AR assembly assistance and monitoring system based on ego-centric vision, *Assem. Autom.* 39 (1) (2019) 1–16, <http://dx.doi.org/10.1108/AA-03-2017-032>.
- [47] Y. Zhang, K. Ding, J. Hui, J. Lv, X. Zhou, P. Zheng, Human-object integrated assembly intention recognition for context-aware human-robot collaborative assembly, *Adv. Eng. Inform.* 54 (2022) 101792, <http://dx.doi.org/10.1016/j.aei.2022.101792>.
- [48] P. Wang, H. Liu, L. Wang, R.X. Gao, Deep learning-based human motion recognition for predictive context-aware human-robot collaboration, *CIRP Ann* 67 (1) (2018) 17–20, <http://dx.doi.org/10.1016/j.cirp.2018.04.066>.
- [49] N.R. Adão Martins, S. Annaheim, C.M. Spengler, R.M. Rossi, Fatigue monitoring through wearables: A state-of-the-art review, *Front. Physiol.* 12 (2021) 790292, <http://dx.doi.org/10.3389/fphys.2021.790292>.
- [50] E.M.A. Smets, B. Garssen, B. Bonke, J. de Haes, The multidimensional fatigue inventory (MFI) psychometric qualities of an instrument to assess fatigue, *Journal Psychosom. Res.* 39 3 (1995) 315–325, [http://dx.doi.org/10.1016/0022-3999\(94\)00125-o](http://dx.doi.org/10.1016/0022-3999(94)00125-o).

- [51] T.H. Bullock, M.C. McClune, J.Z. Achimowicz, V.J. Iragui-Madoz, R.B. Duckrow, S.S. Spencer, Temporal fluctuations in coherence of brain waves, *Proc. Natl. Acad. Sci.* 92 (25) (1995) 11568–11572, <http://dx.doi.org/10.1073/pnas.92.25.11568>.
- [52] A. Dubey, S. Ray, Comparison of tuning properties of gamma and high-gamma power in local field potential (LFP) versus electrocorticogram (ECoG) in visual cortex, *Sci. Rep.* 10 (1) (2020) 5422, <http://dx.doi.org/10.1038/s41598-020-61961-9>.
- [53] S. Jirayucharoensak, S. Pan-Ngum, P. Israsena, EEG-based emotion recognition using deep learning network with principal component based covariate shift adaptation, *Sci. World J.* 2014 (1) (2014) 627892, <http://dx.doi.org/10.1155/2014/627892>.
- [54] M.L. Vicchiotti, F.M. Ramos, L.E. Betting, A.S.L.O. Campanharo, Computational methods of EEG signals analysis for Alzheimer's disease classification, *Sci. Rep.* 13 (1) (2023) 8184, <http://dx.doi.org/10.1038/s41598-023-32664-8>.
- [55] H. Peng, F. Long, C. Ding, Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy, *IEEE Trans. Pattern Anal. Mach. Intell.* 27 (8) (2005) 1226–1238, <http://dx.doi.org/10.1109/TPAMI.2005.159>.
- [56] K.M. Veena, K. Manjula Shenoy, K.B. Ajitha Shenoy, Performance comparison of machine learning classification algorithms, in: M. Singh, P.K. Gupta, V. Tyagi, J. Flusser, T. Ören (Eds.), *Advances in Computing and Data Sciences*, Springer Singapore, Singapore, 2018, pp. 489–497, http://dx.doi.org/10.1007/978-981-13-1813-9_49.
- [57] F. Kong, Z. Lu, L. Kong, T. Chen, Information field in a manufacturing system: Concepts, measurements and applications, *Adv. Eng. Inform.* 56 (2023) 101946, <http://dx.doi.org/10.1016/j.aei.2023.101946>.
- [58] C.E. Shannon, A mathematical theory of communication, *Bell Syst. Tech. J.* 27 (3) (1948) 379–423, <http://dx.doi.org/10.1002/j.1538-7305.1948.tb01338.x>.
- [59] L. Wang, Y. Xiong, Z. Wang, Y. Qiao, D. Lin, X. Tang, L. Van Gool, Temporal segment networks: Towards good practices for deep action recognition, in: *Computer Vision – ECCV 2016*, 2016, pp. 20–36, http://dx.doi.org/10.1007/978-3-319-46484-8_2.
- [60] D. Mutlu-Bayraktar, V. Cosgun, T. Altan, Cognitive load in multimedia learning environments: A systematic review, *Comput. Educ.* 141 (2019) 103618, <http://dx.doi.org/10.1016/j.compedu.2019.103618>.
- [61] R. Saeed, Improving the user experience using an intelligent adaptive user interface in mobile applications, in: *2016 IEEE International Multidisciplinary Conference on Engineering Technology, IMCET*, 2016, pp. 64–68, <http://dx.doi.org/10.1109/IMCET.2016.7777428>.
- [62] S.G. Hart, Nasa-task load index (NASA-TLX); 20 years later, *Proc. Hum. Factors Ergon. Soc. Annu. Meet.* 50 (9) (2006) 904–908, <http://dx.doi.org/10.1177/154193120605000909>.
- [63] J.B. Brooke, *SUS: A 'quick and dirty' usability scale*, 1996.